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 SURVEY

Human Behavior Analysis: A Comprehensive Survey on Techniques, Applications, Challenges, and Future Directions

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ABSTRACT Human Behavior Analysis (HBA) has emerged as a critical interdisciplinary field, combining psychology, sociology, artificial intelligence, and data science to model, understand, and predict human behavior across diverse domains. This paper provides a comprehensive survey, addressing gaps in existing literature by exploring applications, techniques, challenges, and future directions. We begin by defining HBA, tracing its historical roots, and outlining core concepts such as behavioral patterns, cognitive processes, and emotional states. The survey then explores traditional and modern techniques, from manual observation to AI-driven methods such as deep learning, natural language processing, and computer vision. A key contribution is our extensive coverage of HBA applications in healthcare, marketing, education, workplace productivity, activity recognition, and criminal justice. For each domain, we provide detailed examples of how HBA enhances outcomes and decision-making. The survey also delves into data sources and methodologies used in HBA, such as sensor data, social media data, physiological signals, and multimodal analysis. We discuss major challenges such as data privacy, generalization, real-time processing, and scalability. Finally, we highlight emerging trends and future directions, including edge computing, Large Language Models, privacy-preserving techniques, and cross-disciplinary approaches. By offering a holistic review, this survey aims to guide future research and innovation in the evolving field of HBA.

INDEX TERMS Human behavior analysis, human activity recognition, Internet of Things, deep learning, machine learning, computer vision, natural language processing.

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I. INTRODUCTION

Human Behavior Analysis is an interdisciplinary field that seeks to understand, model, and predict human actions,

reactions, and interactions. By analyzing patterns in human behavior, HBA systems aim to provide insights into individual and group activities, mental states, and social dynamics [1]. The study of human behavior is critical across various fields, including healthcare, marketing, education, public safety, and workplace productivity, as it offers a deeper understanding of how individuals engage with their environment and interact with others.

The significance of HBA lies in its ability to translate complex behavioral patterns into actionable insights [2]. In healthcare, for instance, monitoring patient behavior can enable early detection of health problems such as stress, anxiety, or physical ailments. Similarly, in marketing, analyzing consumer behavior provides companies with the ability to tailor products, services, and marketing strategies to better meet consumer needs [3]. In education, HBA can improve learning outcomes by identifying individual learning patterns and providing personalized feedback [4]. The versatility of HBA applications makes it an essential tool in understanding human behavior in modern society.

Historically, the analysis of human behavior relied heavily on qualitative methods, such as interviews, surveys, and observations conducted by psychologists and sociologists [5]. While these methods provided valuable insights into individual behaviors and social trends, they were often limited by the small scale of data and subjective biases. The emergence of digital technologies has transformed HBA into a data-driven discipline. With the proliferation of smartphones [6], wearable devices, and social media, vast amounts of behavioral data are now available, allowing for large-scale, real-time analysis of human activities [7], [8]. This shift has significantly enhanced the accuracy, scalability, and applicability of HBA across different domains.

Modern HBA leverages advanced technologies such as artificial intelligence (AI), machine learning (ML) [10], and sensor networks to analyze various forms of data, including physiological signals [11], facial expressions [12], speech patterns [13], and social media interactions [14]. These technologies enable automated, real-time analysis of human behavior, providing insights that were previously unattainable through manual methods. For example, AI models can now analyze video feeds to detect subtle changes in a person’s facial expressions or body language [15], indicating emotional states such as stress or happiness. Similarly, natural language processing techniques are used to analyze text data, such as social media posts or emails [16], to gauge sentiment and psychological well-being.

One of the key strengths of contemporary HBA is its ability to integrate multiple data sources. By combining data from wearables, video feeds [17], social interactions, and environmental sensors [18], HBA systems create a more comprehensive picture of human behavior. This multimodal approach enables the detection of complex patterns, such as the relationship between physical health, emotional well-being, and social behaviors, offering more nuanced insights

TABLE 1. List of abbreviations.

Abbreviation	Definition
AAS	Automated Attendance Systems
Acc	Accuracy
AE	Auto-Encoder
AI	Artificial Intelligence
ANNs	Artificial Neural Networks
AES	Automated Essay Scoring
BiGRU	Bidirectional Gated Recurrent Unit
BiLSTM	Bidirectional Long Short-Term Memory
CEM	Classroom Engagement Monitoring
CNN	Convolution Neural Network
CV	Computer Vision
CRT	Classification and Regression Tree
DL	Deep Learning
DNN	Deep Neural Networks
DT	Decision Tree
Dims	Dimensions
EEG	Electroencephalogram
EMG	Electromyogram
EWSDP	Early Warning Systems for Dropout Prediction
GAN	Generative Adversarial Network
GA	Genetic Algorithm
GB	Gradient Boosting
GBM	Gradient Boosting Machine
GCN	Graph Convolutional Network
GRU	Gated Recurrent Units
HAR	Human Activity Recognition
HBA	Human Behavior Analysis
HR	Heart Rate
IoT	Internet of Things
ITS	Intelligent Tutoring Systems
KNN	K-Nearest Neighbors
LDA	Linear Discriminant Analysis
LightGBM	Light Gradient Boosting Machine
LR	Logistic Regression
LSTM	Long Short-Term Memory
M	Magnetometer
MAE	Mean Absolute Error
MAP	Mean Average Precision
ML	Machine Learning
MLP	Multi-layer Perceptrons
NB	Naive Bayes
NLP	Natural Language Processing
NN	Neural Network
PCA	Principal Component Analysis
PTTs	Pronunciation Training Tools
RF	Random Forest
RL	Reinforcement Learning
RNNs	Recurrent Neural Networks
SVM	Support Vector Machine
VAER	Variational AutoEncoder Regularization
VAs	Ventricular Arrhythmias
×	Not specified

into human actions [19]. As a result, HBA has moved from being a passive tool for observation to an active system for real-time decision-making and intervention.

A. PAPER MOTIVATION

The motivation for this survey stems from the lack of comprehensive studies that cover the entire scope of HBA. While several surveys exist on Human Activity Recognition (HAR) or Human Behavior Recognition (HBR), these only represent parts of the broader field of HBA. Our survey aims to address this gap by offering a comprehensive review that includes all major application domains of HBA, such as Healthcare,

HAR, Marketing and Consumer Behavior, Education and Learning, Workplace Productivity and Employee Well-being and more. Another significant motivation is the absence of surveys that explore the future trends in HBA, such as the integration of edge computing, Large Language Models and other emerging technologies. Our work seeks to fill these gaps by providing a holistic view of HBA and recommending future research directions.

B. CRITICAL REVIEW OF EXISTING SURVEYS

This section provides a critical review of prior surveys and key research efforts in Human Behavior Analysis, highlighting their scope, methodological frameworks, and limitations in order to contextualize the contributions of the present work. Below, we provide a brief summary of the most significant related works and how they contribute to the current understanding of HBA (see Table 2).

Bruno et al. [20] provided a comprehensive overview of advancements in human action recognition, highlighting the transition from handcrafted features to deep learning techniques. It categorizes the problem into action classification and spatiotemporal action localization, discusses current challenges such as inference time and video annotation complexities, and emphasizes the need for unified architectures and semi-supervised learning approaches. The survey serves as a valuable resource for understanding the state-of-the-art methods and future directions in the field of HBA.

Andrew et al. [21] focused on modeling, replicating, and predicting human behavior in artificial systems, highlighting techniques such as reinforcement learning to learn behavior models through exploration and feedback, as well as methods for modeling human reasoning processes such as beliefs and biases. The key contributions included a detailed analysis of how AI agents could better interact with humans by learning from human actions and understanding mental states, which enabled human-AI hybrid systems to operate more effectively together. The survey also covered applications in autonomous systems, human-AI interaction, and adaptive learning techniques, providing insights into future developments in human-centric AI.

Alejandro et al. [22] provided a comprehensive overview of the systematic identification, analysis, and categorization of relevant works in the field of User Behavior Analysis (UBA) across domains such as cybersecurity, networks, safety and health, and service delivery improvement. It introduced a global relevance score for ranking papers and provided a graphical visualization based on similarity metrics, offering insights and guidelines for researchers interested in UBA. The survey also highlighted the strong and weak points of existing approaches and identified open challenges and future research directions.

Gowsikhaa et al. [23] provided a comprehensive overview of automated HBA from surveillance videos, focusing on methods for detecting, classifying, and tracking abnormal activities for enhanced security. It emphasized the shift from manual to automated surveillance, highlighting key

techniques such as motion detection, object classification, and motion tracking, which form the building blocks of behavior analysis. The survey also explored the integration of semantic analysis to improve accuracy and domain-specific event recognition, while addressing the challenges and limitations in areas such as occlusion handling, person identification, and scene classification.

Unlike previous surveys that primarily catalog algorithms or focus on narrow subdomains such as Human Activity Recognition or behavior modeling in isolated contexts, this review is designed to offer a holistic and critical synthesis of the HBA landscape. Our goal is not merely to summarize existing techniques, but to analyze methodological patterns, highlight domain-specific challenges, and identify overarching limitations across a wide range of application areas—including healthcare, education, marketing, public safety, workplace well-being, and financial decision-making. By comparing models, evaluation approaches, data sources, and deployment barriers, we aim to surface research gaps and actionable insights that can inform future work. This survey also integrates cross-domain discussions, ethical considerations, and emerging trends such as edge computing and privacy-preserving AI to ensure a forward-looking perspective. In doing so, we aim to position this work not just as a repository of knowledge, but as a roadmap for advancing the field of HBA in both theory and practice.

C. MAIN CONTRIBUTIONS

The main contributions of this survey are summarized as follows:

- The paper provides a comprehensive review of the field of HBA, covering its definition, core concepts (behavioral patterns, cognitive processes, emotional states), and historical context, with a focus on modern AI-based methods.
- The survey systematically explores a wide range of application domains of HBA, including healthcare and well-being, marketing and consumer behavior, activity recognition, education and learning, workplace productivity and employee well-being, and criminal justice and public safety.
- The paper presents a detailed review of modern methodologies in HBA, including ML, DL, NLP, and computer vision, showing how these advanced techniques have transformed behavioral analysis.
- The survey offers a comprehensive analysis of the data sources used in HBA, such as sensor data, social media data, and physiological signals. It also discusses various data collection techniques such as wearable devices, IoT sensors, and mobile apps. It discusses also benchmark datasets for HBA Applications.
- A critical examination of the challenges and limitations in HBA is provided, addressing issues such as data privacy, generalization and bias in models, real-time processing, and scalability.

TABLE 2. Critical comparison of existing surveys and our work.

Ref.	Survey Type	HBA Background	Application Domains	Techniques	Challenges	Future Directions	Cross-Domain Analysis	Summary
[22]	Systematic	✓	✓	✓	✗	✗	✗	Broad coverage of user behavior analysis across multiple sectors; lacks critical synthesis and future-focused perspective.
[21]	Comprehensive	✓	✗	✓	✓	✓	✗	Explores behavioral modeling and AI-human interaction in depth; lacks domain-level diversity and cross-sector analysis.
[23]	Systematic	✓	✓	✓	✗	✗	✗	Focuses on surveillance and abnormal behavior in videos; limited scope and no exploration of broader applications or technologies.
[20]	Comprehensive	✓	✓	✓	✗	✗	✗	Focuses on action recognition and deep learning evolution; lacks coverage of applications beyond video and limited future analysis.
Our Survey	Comprehensive	✓	✓	✓	✓	✓	✓	Broad, critical synthesis of HBA methods and applications across multiple domains including healthcare, marketing, education, and workplace. Incorporates methodological critique, ethical considerations, and forward-looking research gaps.

- The paper identifies emerging trends and future directions in HBA, including the integration of edge computing, Large Language Models, AI-powered behavioral models, and advanced sensing technologies. The survey also highlights cross-disciplinary approaches and the role of privacy-preserving techniques such as federated learning and differential privacy.
- The survey highlights the importance of ethical considerations in HBA, particularly recommending privacy-preserving methodologies. It stresses the necessity for future advancements in the field to align with ethical standards.

D. PAPER ORGANIZATION

The structure of this survey is organized as follows (see Figure 1): Section I introduces the background and motivation for HBA. Section III provides an overview of HBA, including its key concepts, historical context, and modern techniques. Section IV discusses the data sources and methodologies employed in HBA. Section V reviews the various application domains of HBA. Section VI addresses the challenges and limitations in HBA, including data privacy and scalability. Section VII explores future trends and emerging technologies in HBA. Finally, Section VIII concludes with a summary of key findings and implications for future research.

II. REVIEW METHODOLOGY

In this section, we present the review methodology employed to explore the current landscape of HBA. A systematic and structured approach was adopted to ensure comprehensive coverage of the literature, eliminate bias, and support the critical synthesis of trends, technologies, and application domains. This section details the search strategy, inclusion and exclusion criteria, and the analysis methods used to derive insights from the selected body of literature.

A. SEARCH STRATEGY

To identify the relevant literature for this survey on HBA, a comprehensive keyword-based search strategy was

employed using terms such as “Human Behavior Analysis”, “Behavioral Modeling”, “Behavior Prediction”, “Human Behavior Recognition”, “Affective Computing”, “Emotion Recognition”, “AI for Behavior Understanding”, and “Machine Learning for Behavior Modeling”. Additionally, application-specific terms were incorporated to capture the breadth of cross-domain relevance. These included “HBA in Healthcare”, “HBA in Marketing”, “Consumer Behavior Prediction”, “Behavior Analysis in Education”, “Behavior Monitoring in Learning Environments”, “Workplace Productivity Prediction”, “Mental Health and Stress Detection”, “Crime Prediction using AI”, “Public Safety and Behavioral Analysis”, “Driver Behavior Prediction”, “Human Activity Recognition”, and “Behavioral Finance Analysis”. The search was conducted across multiple leading academic databases, including IEEE Xplore, ACM Digital Library, Scopus, Web of Science, SpringerLink, ScienceDirect, and Wiley Online Library. Only peer-reviewed journal articles, conference proceedings, and high-quality technical reports published between 2020 and 2024 were considered to ensure relevance to recent advancements. The focus was on studies that applied AI, ML, or DL techniques to analyze, predict, or model human behavior in diverse domains such as healthcare, education, workplace environments, marketing, transportation, and public safety and more.

B. INCLUSION/EXCLUSION CRITERIA

Table 3 presents the inclusion and exclusion criteria applied in our study selection process to ensure the relevance, quality, and scientific rigor of the surveyed papers. The inclusion criteria emphasize research that directly applies ML, DL, or AI techniques to HBA across multiple domains, including but not limited to healthcare, education, transportation, workplace productivity, and public safety. Studies with theoretical models, empirical evaluations, or real-world applications were considered. Exclusion criteria were set to filter out non-relevant, outdated, or methodologically weak papers that do not directly contribute to understanding or advancing the HBA field.

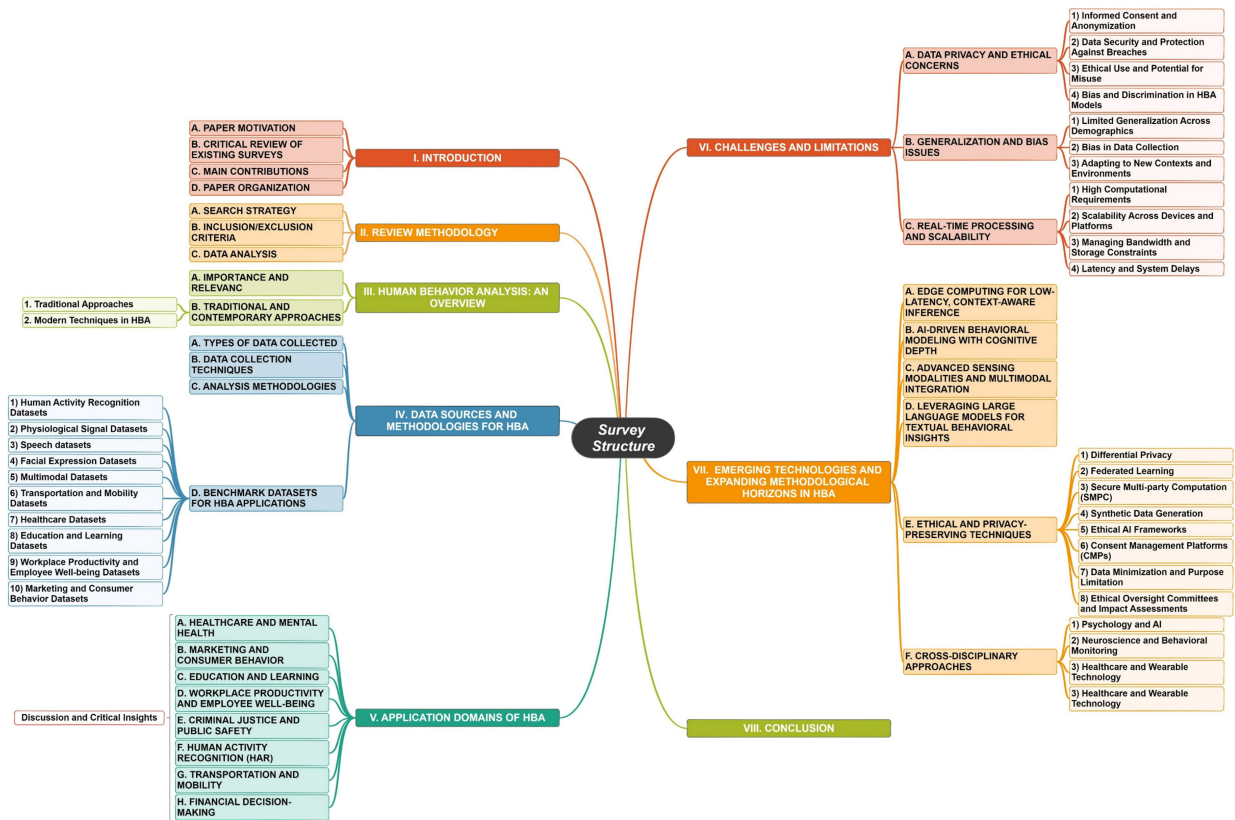


FIGURE 1. Paper organization.

C. DATA ANALYSIS

Following the search and selection process, a total of 202 relevant papers were identified and analyzed across eight core domains of HBA (see Figure 2). These domains include healthcare and mental health, marketing and consumer behavior, education and learning, workplace productivity and employee well-being, criminal justice and public safety, human activity recognition, transportation and mobility, and financial decision-making.

Out of the 202 papers, 134 were journal articles and 61 were conference proceedings, with an additional 7 being categorized as preprints. No book chapters, books, or monographs were included in the final selection. Human Activity Recognition emerged as the most represented area with 47 papers (33 articles and 14 proceedings), highlighting the maturity and depth of research in this subdomain. Criminal Justice and Public Safety followed with 36 papers (24 articles, 5 proceedings, and 7 preprints), reflecting a growing interest in behavior modeling in public safety environments.

Education and learning contributed 26 papers (24 articles and 2 proceedings), while Transportation and Mobility accounted for 20 papers (15 articles and 5 proceedings). Both Healthcare and Mental Health and Workplace Productivity and Employee Well-being yielded 18 papers each. However, workplace productivity research was almost entirely repre-

sented in proceedings (17 out of 18), indicating a prevalence of recent, possibly ongoing studies.

Marketing and Consumer Behavior presented 17 papers (13 articles and 4 proceedings), focusing on behavioral analysis in commercial settings. Lastly, Financial Decision-Making included 20 papers (7 articles and 13 proceedings), illustrating the integration of behavioral modeling into economic systems and risk management.

This analysis highlights not only the cross-disciplinary nature of HBA but also the differing levels of maturity and dissemination across domains. It offers a robust foundation for synthesizing methodological patterns, evaluating application-specific insights, and identifying areas requiring further scholarly attention.

III. HUMAN BEHAVIOR ANALYSIS: AN OVERVIEW

HBA refers to the process of studying human actions, reactions, cognitive processes, and emotional states in different environments (see Figure 3). The core concepts behind HBA include:

- *Behavioral Patterns* [24]: Refers to the regular and predictable actions exhibited by individuals or groups over time. These patterns are often detected through observation, data collection, and analysis of behaviors

TABLE 3. Inclusion/exclusion criteria for selecting studies on HBA.

Selection Criteria	Description
Inclusion Criteria:	
• Studies that explore Human Behavior Analysis using AI, ML, or DL.	• To ensure the relevance of methodologies and contributions to HBA research.
• Peer-reviewed articles, high-impact conference papers, and authoritative technical reports.	• To maintain academic integrity and reliability of findings.
• Papers published between 2020 and 2024.	• To focus on recent innovations and emerging trends in HBA.
• Studies covering real-world applications, empirical evaluations, or system-level implementations.	• To prioritize practical relevance and validated contributions.
• Research spanning cross-domain applications (e.g., healthcare, marketing, education, transportation, finance, public safety).	• To cover the full landscape of modern HBA applications.
Exclusion Criteria:	
• Studies unrelated to behavioral modeling, prediction, or recognition.	• To remove papers that do not address HBA directly.
• Articles lacking experimental validation or meaningful theoretical insight.	• To exclude speculative or anecdotal contributions without empirical support.
• Non-English publications.	• To maintain accessibility and consistency in language.
• Duplicate studies or reviews without significant differentiation.	• To eliminate redundancy and focus on unique contributions.
• Outdated reviews or works limited to legacy technologies (pre-2020 without recent updates).	• To focus on current state-of-the-art and active research directions.

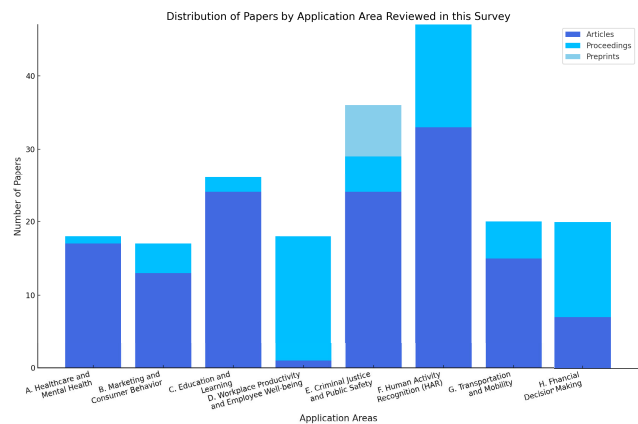


FIGURE 2. Distribution of papers by application area reviewed in this survey.

in specific contexts, such as social interactions, online activities, or physical movements.

- *Cognitive Processes* [25]: These are mental processes involved in gaining knowledge and understanding, including memory, attention, perception, and decision-making. HBA examines how these processes influence observable behaviors, such as responses to stimuli or problem-solving approaches.
- *Emotional States* [26]: Human emotions play a critical role in influencing behavior. Emotions such as happiness, sadness, fear, or anger can be analyzed through

facial expressions, speech, and physiological responses, providing valuable insights into underlying motivations and actions.

The study of human behavior has evolved significantly from its origins in fields such as psychology and sociology to modern data-driven techniques. Historically, psychology-based approaches [27] relied on observational studies, surveys, and experiments to understand individual and group behavior. These methods were often limited by subjective interpretation, small sample sizes, and a reliance on self-reported data.

In parallel, sociology-based approaches [28] examined human behavior in societal contexts, focusing on social structures, norms, and group dynamics. These early methods laid the groundwork for understanding human behavior in relation to the environment, but they lacked the computational power to process large-scale behavioral data.

With the advent of AI and ML, the field of HBA underwent a transformation. AI-powered approaches enable the analysis of vast datasets, capturing behavioral nuances that were previously difficult to observe. For example, facial recognition technologies can identify micro-expressions, speech analysis can detect emotional tone, and wearable sensors can monitor physiological signals such as heart rate or movement patterns.

Modern HBA techniques use a combination of big data, NLP, and DL to automate the analysis of human behavior. This shift has enabled real-time analysis, scalable models,

and more objective, data-driven insights, making HBA applicable to various industries such as healthcare, marketing, education, and security.

A. IMPORTANCE AND RELEVANCE

HBA has become increasingly important in today's interconnected, technology-driven world. Its relevance can be seen in a variety of real-world applications:

- *Healthcare Monitoring*: HBA is used for monitoring patients' behaviors, detecting abnormalities, and improving treatment plans. For example, wearable devices track physical activity, sleep patterns, and heart rates, which can help in managing chronic conditions or detecting early signs of illness. Mental health monitoring through HBA helps identify early signs of stress, anxiety, or depression by analyzing speech patterns, social media activity, and physiological indicators.
- *Smart Environments*: HBA is crucial for creating adaptive systems that respond to user behaviors. For instance, smart homes use behavioral data to adjust lighting, temperature, and security settings automatically based on residents' preferences and routines. In transportation, HBA is applied in autonomous vehicles, traffic management, and public transportation systems to optimize routes, reduce congestion, and improve safety.
- *Security*: HBA is integral to enhancing security systems, both in physical and digital spaces. By analyzing behaviors such as gait, facial expressions, or online activity, systems can detect anomalies that indicate potential security threats. In cybersecurity, HBA helps in identifying malicious behavior patterns such as phishing attempts, unauthorized access, or fraudulent transactions by analyzing user activity and network traffic.
- *Education*: HBA is employed to enhance learning experiences by understanding how students interact with educational materials and platforms. Adaptive learning systems use behavior data to personalize instruction, while engagement monitoring tools track student participation and provide real-time feedback to educators.
- *Marketing and Consumer Behavior*: Companies leverage HBA to understand consumer preferences and tailor marketing strategies. Behavioral data, including purchase history, website interactions, and social media activity, are analyzed to create personalized recommendations, targeted advertisements, and improved customer service experiences.

B. TRADITIONAL AND CONTEMPORARY APPROACHES

The study of human behavior has evolved from traditional methods involving manual data collection and psychological evaluation to modern approaches powered by advanced computational techniques. Traditional approaches were often limited in scope and scale, while contemporary methods leverage AI and ML to analyze large datasets and generate

real-time insights. This section outlines both traditional and modern techniques used in HBA and provides a comparative analysis.

1) TRADITIONAL APPROACHES

Traditional approaches to HBA relied heavily on manual data collection, observation, and subjective analysis. These methods were often used in fields such as psychology, sociology, and anthropology, focusing on understanding individual and group behavior within specific environments. Key traditional techniques include:

- *Manual Observations* [29]: Researchers observed subjects in natural or controlled environments, noting behavioral patterns, reactions, and social interactions. These observations were often subjective and dependent on the skill of the observer.
- *Psychological Evaluations* [30]: Psychological assessments, such as personality tests, cognitive evaluations, and self-reports, were widely used to understand behavior from a mental health perspective. While effective for certain analyses, these methods often suffered from biases and were limited to small sample sizes.
- *Survey-Based Data Collection* [31]: Surveys and questionnaires were used to gather information on attitudes, beliefs, and behaviors. This method allowed for data collection from larger populations but was often limited by self-report bias and the reliability of responses.
- *Ethnographic Studies* [32]: Long-term observational studies where researchers embedded themselves in the culture or environment they were studying to understand human behavior in context. These studies provided rich qualitative insights but were time-consuming and difficult to scale.

While these traditional methods provided foundational knowledge of human behavior, they were often limited by their reliance on subjective interpretation, smaller datasets, and manual processing, which hindered scalability and real-time application.

2) MODERN TECHNIQUES IN HBA

With advancements in AI, ML, and data science, HBA has undergone a transformation. Modern approaches allow for the collection, analysis, and interpretation of vast amounts of behavioral data in real-time, improving both the scalability and accuracy of HBA. Key modern techniques include:

- *Deep Learning (DL)*: DL models, particularly CNNs and RNNs, are used to analyze complex data patterns. In HBA, DL is applied to tasks such as emotion detection, facial recognition, and speech analysis, enabling real-time behavior monitoring [33].
- *Natural Language Processing (NLP)*: NLP techniques are used to analyze textual and speech data, providing insights into emotions, intentions, and cognitive states [34]. For example, sentiment analysis is used

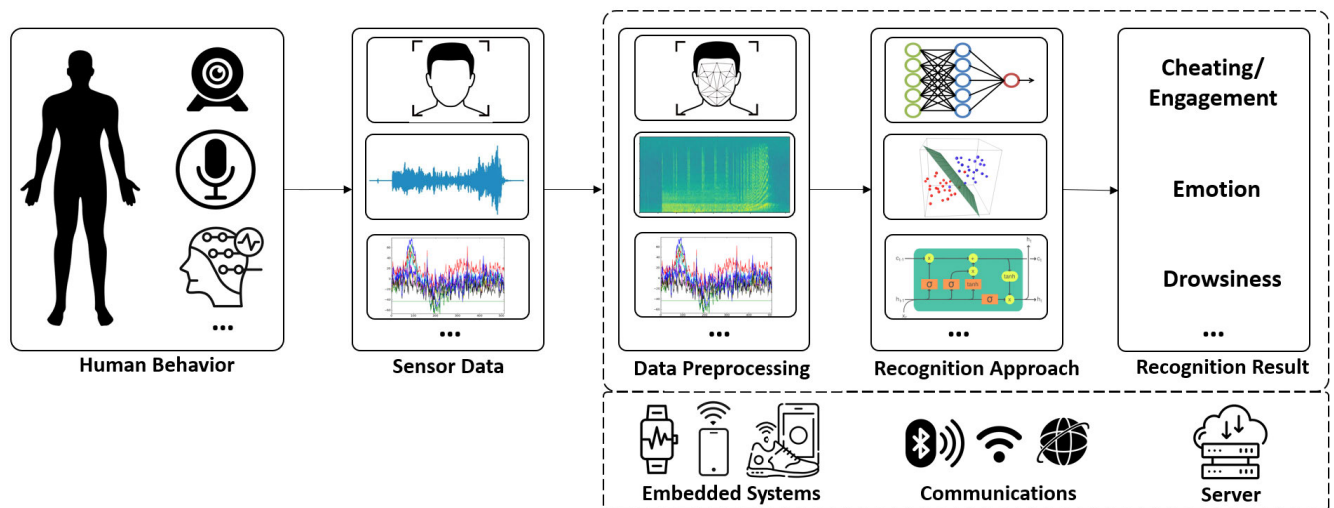


FIGURE 3. The HBA workflow from data collection to behavior recognition.

to detect mood changes in social media posts, while chatbots utilize NLP to respond based on user input.

- **Computer Vision:** Computer vision techniques, particularly those using DL, enable the automatic detection of human actions, facial expressions, and movements [35]. This technology is widely used in security systems, healthcare monitoring, and human-computer interaction to analyze visual behavioral data.
- **Sensor Data Analysis [36]:** Wearable sensors (e.g., heart rate monitors, accelerometers, and GPS devices) collect real-time physiological and movement data. These sensors are used to monitor health conditions, physical activity, and location-based behaviors, providing continuous data that can be analyzed for behavior patterns.
- **Reinforcement Learning:** This technique is used to train models that can predict and adapt to human behavior over time [37]. For instance, in personalized learning environments, systems use reinforcement learning to adapt content based on how users interact with the material.

Modern techniques are data-driven, capable of processing vast amounts of information, and offer real-time insights into human behavior. These approaches have been applied across various fields, from healthcare and smart cities to cybersecurity and consumer behavior analysis.

IV. DATA SOURCES AND METHODOLOGIES FOR HBA

HBA relies on a variety of data sources and methodologies to capture, analyze, and interpret human actions and cognitive processes. This section explores the types of data used in HBA, the techniques for collecting behavior-related data, and the methodologies applied to analyze the collected

data. Additionally, it highlights the challenges faced in data collection and analysis.

A. TYPES OF DATA COLLECTED

The data used in HBA comes from diverse sources, capturing both physical and digital interactions of individuals. These data types include:

- **Sensor Data:** Collected from wearable devices (e.g., smartwatches, fitness trackers), environmental sensors, or IoT devices [36]. These data capture physical activities (e.g., walking, heart rate) and environmental factors (e.g., temperature, light levels).
- **Behavioral Logs:** Behavioral logs capture interactions between individuals and systems, such as clickstreams on websites, app usage, and human-computer interaction logs [38]. These are often used to study online behavior, preferences, and engagement.
- **Social Media Data:** Data from platforms such as Twitter, Facebook, and Instagram are analyzed to understand emotional states, opinions, and social interactions [39]. Sentiment analysis is commonly applied to this type of data to detect mood changes or trends.
- **Physiological Signals [11]:** These include heart rate, skin conductance, EEG (brain activity), and EMG (muscle activity), collected through wearable devices and specialized medical equipment. This data type is essential for understanding emotional and cognitive states in real time.
- **Audio and Video Data:** Audio recordings of speech, video footage of human activity, and facial expression data are used for detecting emotions, analyzing movements, and monitoring behavioral patterns [40].

These data types provide a comprehensive picture of human behavior, allowing researchers to model actions, cognitive processes, and emotional responses.

B. DATA COLLECTION TECHNIQUES

Collecting data for HBA involves a variety of techniques that depend on the context and the specific behaviors being studied. Key data collection methods include:

- *Wearable Devices* [41]: Smartwatches, fitness trackers, and biosensors collect real-time data on physical activity, heart rate, sleep patterns, and more. Wearables provide continuous monitoring and are particularly useful in healthcare and fitness applications.
- *IoT Sensors* [36]: IoT devices, such as smart home systems, environmental sensors, and traffic monitoring cameras, collect data on movement, temperature, light, and location. These sensors enable the monitoring of human behavior in both personal and public spaces.
- *Mobile Apps*: Applications on smartphones or tablets track user interactions, location data, and app usage behavior. Mobile apps are effective for capturing a wide range of user data, including GPS location, accelerometer data, and online behavior [42].
- *Web Platforms and Social Media*: Web platforms and social media sites provide data on user interactions, sentiments, and preferences. Tools such as browser extensions, tracking pixels, and APIs collect data from users interacting with websites and social networks [43].

These collection techniques provide rich, multidimensional datasets that support advanced analysis for understanding human behavior in various contexts.

C. ANALYSIS METHODOLOGIES

Once data is collected, various methodologies are applied to analyze behavioral patterns and cognitive states. The following are key methodologies used in HBA:

- *Time-Series Analysis*: Time-series methods analyze data collected over time, identifying trends, patterns, and anomalies in behavior. For example, time-series analysis is used to study sleep patterns, physical activity, and online engagement across different periods.
- *Behavior Modeling*: Behavior modeling techniques, such as Markov models and agent-based models, simulate human behavior in different scenarios. These models help predict future actions based on past behaviors and environmental conditions.
- *ML Algorithms*: ML techniques, including supervised and unsupervised learning, are used to classify behaviors, predict outcomes, and identify patterns in large datasets. Algorithms such as SVM, DT, RF, and NN are commonly applied.
- *Multimodal Analysis*: Multimodal analysis combines data from multiple sources (e.g., sensor data, video, audio) to provide a more holistic view of human behavior. For example, combining video analysis with physiological signals can enhance emotion detection and cognitive state assessment.

These methodologies enable deeper insights into human behavior, providing both descriptive and predictive analytics in HBA applications.

D. BENCHMARK DATASETS FOR HBA APPLICATIONS

The following section presents a comprehensive overview of key benchmark datasets that are widely utilized in HBA tasks, covering a range of domains including HAR, sentiment analysis, transportation, healthcare, and more.

1) HUMAN ACTIVITY RECOGNITION DATASETS

Numerous benchmark datasets support the development and evaluation of machine learning models in human activity recognition. The WISDM dataset [44] contains over 1.1 million accelerometer readings from 36 users and is commonly used to classify motion patterns from typical physical activities. The UCI-HAR dataset [45], collected from 30 users using smartphone accelerometers and gyroscopes, includes 10,299 labeled instances across six activities and serves as a standard baseline for activity recognition. USC-HAD [46] provides data from 14 subjects performing 12 activities over a period of 7 hours, while PAMAP2 [47] includes 18 labeled physical activities from 9 participants recorded through multiple wearable sensors for multimodal recognition and health monitoring. The Daphnet FoG dataset [48] focuses on detecting freezing-of-gait episodes in Parkinson's disease patients using high-frequency accelerometer data. The MHEALTH dataset [51] is another health-related dataset capturing physical movements and vital signs from 10 individuals using wearable sensors. In industrial environments, the Skoda dataset [49] logs over 60,000 annotated arm movements from sensors embedded in workers' arms during assembly tasks. For video-based recognition, the KTH dataset [50] includes 600 video sequences of six actions performed by 25 subjects under four different scenarios, and the UCF101 dataset [55] offers over 13,000 video clips sourced from YouTube across 101 action categories. Kinect-based datasets such as KARD [52] and PKU-MMD [53] include skeletal joint and depth data across a wide range of activities, supporting gesture and interaction recognition. The OPPORTUNITY dataset [54] provides multimodal recordings from wearable sensors capturing 242 data streams across locomotion and gestures. Finally, the LIRIS dataset [56] presents video sequences of 10 different activities recorded in realistic indoor settings, offering diverse scenarios for activity classification and recognition.

2) PHYSIOLOGICAL SIGNAL DATASETS

Several benchmark datasets capture physiological signals relevant to HBA, especially for emotion and stress recognition. The CLAS dataset [57] contains PPG, ECG, and EDA recordings from 62 participants who engaged in tasks involving emotive imagery and video clips, with annotated labels for stress, valence, and arousal. The DeepBreath dataset [58] focuses on breathing patterns captured via

thermal imaging near the nostrils, using a low-cost thermal camera to assess psychological stress; data were collected from eight participants over sessions lasting 63 to 72 minutes. The ASCERTAIN dataset [59] offers multimodal physiological responses, including EEG, ECG, and GSR, along with facial activity captured via webcam while 58 participants viewed emotionally stimulating film clips. These datasets support the development of affective computing models and stress-detection systems using wearable or non-intrusive sensors.

3) SPEECH DATASETS

In the domain of speech-based emotion and stress analysis, the SUSAS dataset [60] is widely used. It includes speech recordings collected under stress across different speaking styles and environmental conditions. The dataset features 35 potentially confusing aviation-related terms and consists of 16,000 utterances from 32 speakers, making it suitable for modeling stress detection and emotion recognition from vocal expressions.

4) FACIAL EXPRESSION DATASETS

Facial expression datasets are widely used for training and evaluating models in emotion recognition and stress detection. The CK+ dataset [61] includes 593 image sequences from 123 participants, annotated for both facial expressions and action units. The Oulu-CASIA dataset [62] provides facial expression data from 80 individuals, covering six basic emotions captured under three different lighting conditions. The KDEF dataset [63] consists of 4,900 images featuring 70 individuals expressing a range of emotional states. For real-world applications, the KMU-FED dataset [64] captures 55 sequences from 12 subjects in actual driving scenarios using near-infrared cameras mounted inside vehicles. The FERET dataset [65], originally created for facial recognition research, contains 14,126 facial images from 1,199 individuals collected over 15 sessions between 1993 and 1996. Lastly, the ANUStressDB dataset [66] features thermal and visible-spectrum video recordings of 35 individuals watching emotionally evocative films, with annotations distinguishing stressful from non-stressful responses.

5) MULTIMODAL DATASETS

Multimodal datasets provide rich, synchronized data streams from multiple sources such as audio, video, and physiological signals, enabling comprehensive behavioral analysis. The NNIME dataset [67] includes synchronized audio, video, and ECG recordings from 44 professionally trained actors participating in 102 dyadic interaction sessions. The dataset spans approximately 11 hours of annotated recordings across six emotional categories, supporting research in emotion recognition and multimodal interaction. The TILES-2018 dataset [68] offers large-scale real-world data from 212 hospital staff in California, capturing continuous ECG, respiration, physical activity, speech-derived features, and other

sensor-based metrics through wearable devices for behavioral and wellness monitoring in workplace environments.

6) TRANSPORTATION AND MOBILITY DATASETS

Transportation and mobility datasets enable the analysis of driver behavior, distraction, fatigue, and stress through multimodal signals and in-vehicle contexts. The State Farm Distracted Driver Detection dataset [69] includes over 22,000 labeled dashboard camera images of drivers in various distracted states, used for classifying visual activities. The Drive&Act dataset [70] provides 13 hours of data from 15 subjects, covering over 83 in-car activities through RGB, depth, thermal, 3D skeleton, and synchronized IMU signals. The Honda Research Institute Driving Dataset (HDD) [76] contains over 100 hours of real-world driving videos with GPS, IMU, and CAN bus data, annotated for high-level driving actions and driver behaviors. Several datasets focus on physiological and cognitive monitoring during driving. The SRAD dataset [73] captures four types of physiological signals during driving sessions to assess driver stress. The SEED-VIG dataset [71] collects EEG data to evaluate vigilance states like drowsiness and alertness. The MPDB dataset [77] integrates EEG, ECG, EMG, GSR, and eye movement data from 35 participants in a driving simulator to classify five categories of driving behavior. Similarly, the MDVFDD dataset [72] combines physiological, facial, eye-tracking, and vehicle control data from simulated driving to study attention and fatigue. The DMD dataset [75] offers 41 hours of multimodal recordings from 37 drivers for distraction and behavior analysis. The ADARP dataset [74] targets mental health and stress detection in drivers with alcohol use disorder by collecting HR, EDA, temperature, and movement signals. These datasets collectively support research in autonomous driving, driver monitoring, human factors, and behavioral modeling in transport contexts.

7) HEALTHCARE DATASETS

Healthcare datasets provide critical physiological and clinical data for developing models in disease diagnosis, monitoring, and behavioral analysis. The CHFDB dataset [78] includes long-duration ECG recordings (around 20 hours) from 15 patients with severe congestive heart failure and is widely used in cardiac research and arrhythmia detection. The MIT-BIH dataset [79] offers 48 half-hour ECG segments capturing various arrhythmias and normal rhythms, serving as a foundational dataset in cardiology and machine learning studies. Similarly, the PTB Diagnostic ECG dataset [84] provides 549 ECG recordings from 290 individuals, including those with myocardial infarction and heart failure. For sleep disorder analysis, the Apnea-ECG dataset [85] includes annotated ECG recordings from 70 individuals, distinguishing between apneic and non-apneic patterns. The CHB-MIT Scalp EEG dataset [80] comprises EEG recordings from 22 pediatric epilepsy patients, including seizure and non-seizure events, and is commonly used in seizure detection

research. For fetal health monitoring, the Abdominal and Direct Fetal ECG dataset [81] captures ECG signals from pregnant women via abdominal and fetal electrodes. The BCI IV 2a dataset [83] provides EEG recordings from nine subjects performing motor imagery tasks, useful in brain-computer interface studies. The MIMIC-IV dataset [82] is a large-scale clinical database with detailed ICU records of over 40,000 patients, including vital signs, lab results, diagnoses, and treatments. Lastly, the Electronic Nose Dataset for COPD [86] contains breath samples from smokers and healthy individuals, collected via sensors detecting volatile organic compounds, aimed at distinguishing COPD cases from healthy controls.

8) EDUCATION AND LEARNING DATASETS

Education-focused datasets offer multimodal insights into student behavior, engagement, and academic performance. The StudentLife dataset [87] captures passive sensing data from mobile phones over 10 weeks from 48 Dartmouth students to study mental health, behavior, and academic outcomes. The ATL-BP dataset [88] includes 2,749 labeled video samples of 54 students interacting with an online tutor, providing facial and gesture data for engagement and affective analysis. The Automated Online Exam Proctoring dataset [89] contains multimedia inputs such as webcam and microphone recordings to detect cheating behaviors in real-time, using gaze tracking, user verification, and phone usage detection. The Student Performance dataset [90] was collected from 326 university students in Oman and includes 40 features across academic records, Moodle activity logs, and video interactions. The xAPI-Educational Mining dataset [91] follows the Experience API (xAPI) format and logs detailed student interactions in a virtual learning environment, supporting behavioral pattern analysis. Lastly, the SCB-dataset [92] provides 4,003 classroom images and over 11,000 labeled instances, specifically targeting hand-raising behavior for automatic student behavior detection in real-world classroom settings.

9) WORKPLACE PRODUCTIVITY AND EMPLOYEE WELL-BEING DATASETS

Datasets related to workplace productivity and well-being enable the analysis of physiological stress, mental health, and organizational performance. The WESAD dataset [93] collects physiological signals such as heart rate, respiration, and temperature using both wrist- and chest-mounted sensors, capturing stress and relaxation states including meditation. Similarly, the S-Test dataset [94] includes heart rate, EDA, skin temperature, and contextual factors like activity and ambient conditions to assess wearable-based stress detection. The DS-3 dataset [95] replicates the Trier Social Stress Test (TSST) digitally, providing real-time physiological recordings that illustrate individuals' stress responses during and after stressful tasks. For HR analytics, the Employee's Performance dataset [96] includes organizational data such as

performance indicators and appraisals, aiding in understanding engagement and productivity trends. Additionally, the HR Employee Attrition and Performance dataset [97] captures information on employee demographics, satisfaction, education, and work-life balance to support retention analysis and identify predictors of attrition within organizations.

10) MARKETING AND CONSUMER BEHAVIOR DATASETS

Datasets in marketing and consumer behavior offer valuable insights into purchasing patterns, financial sentiment, and customer engagement. The Online Retail dataset [98] contains transactional records from a UK-based retailer from 2010 to 2011, including invoice details, stock codes, and country data, widely used for consumer segmentation and market basket analysis. The Instacart Market Basket dataset [99] features over 3 million anonymized grocery orders from more than 200,000 users, providing detailed information on shopping habits, order frequency, and product categories. The Bank Marketing dataset [100] includes responses from 41,188 clients contacted in a telemarketing campaign to promote term deposits, along with demographic and interaction data, making it ideal for targeted campaign modeling. The UK Consumer Trends dataset [101] provides long-term expenditure statistics from 1997 to 2022 across diverse sectors such as food, health, housing, and education, supporting economic behavior analysis. Lastly, the Consumer Complaint dataset [102] contains records of financial grievances related to loans, credit, and money transfers, which are frequently used to train models for complaint classification and regulatory insight.

Table 4 provides a comprehensive overview of benchmark datasets for HBA categorized by domain.

V. APPLICATION DOMAINS OF HBA

In this section, we will explore various applications of HBA using modern AI-based methods (see Figure 4).

A. HEALTHCARE AND MENTAL HEALTH

Several studies have used logistic regression (LR) for health-care predictions, including ICU admissions and in-hospital mortality forecasting [103], as well as early warning systems for health crises when combined with GBM [104]. SVM, ANN, RF, and KNN were applied in the diagnosis of chronic kidney disease [105], survival assessment models for heart failure [106], and predicting clinical outcomes for various health conditions, showing an accuracy of 85% with RF, SVM, and LR [107], both showing high accuracy. LSTM models were utilized for real-time health and fitness monitoring, effectively recognizing activities and health zones [108].

In neurological diseases, ensemble learning techniques, such as combining MFA, MMD, and MRD, were used to improve Alzheimer's disease classification [109], while VAER paired with K-Means clustering was employed for real-time detection of Parkinson's disease [110]. IoT-based

TABLE 4. Benchmark datasets for HBA applications.

Domain	Dataset	Ref.	Year	Number of Samples	Data Type	Applications
Human Activity Recognition	KTH	[50]	2004	600 video sequences	Video data	Video-based HAR
	Skoda	[49]	2008	60,000 instances	A	Industrial activity recognition
	OPPORTUNITY	[54]	2010	242 signals	A, G, M	Context awareness, sensor fusion
	Daphnet FoG	[48]	2010	Data from 10 patients	A	Freezing of gait detection in Parkinson's
	WISDM	[44]	2011	1.1 million instances	Smartphone A	HAR
	USC-HAD	[46]	2012	7 hours of data	A, G	Activity recognition
	PAMAP2	[47]	2012	10 hours of data	IMUs, HR monitor	Physical activity recognition
	UCF101	[55]	2012	13,320 video clips	Video data	Action recognition
	UCI-HAR	[45]	2013	10,299 instances	A, G	HAR
	LIRIS	[56]	2014	×	Video data	Complex human activity recognition
Sentiment Analysis	MHEALTH	[51]	2014	×	A, G, M, ECG	Healthcare applications
	KARD	[52]	2017	×	Microsoft Kinect	Gesture and activity recognition
	PKU-MMD	[53]	2017	1,000+ video sequences	RGB, depth, 3D skeleton data	Multimodal action recognition
	The FERET	[65]	1996	14,126 photos	Facial photographs	Facial recognition algorithm development
	SUSAS	[60]	1997	16,000 utterances	Speech data	Speech under stress recognition
	KDEF	[63]	1998	4,900 images	Images of facial expressions	Facial expression recognition
	CK+	[61]	2010	593 picture sequences	Facial expression images	Facial emotion recognition
	Oulu-CASIA	[62]	2011	80 individuals	Images of facial expressions	Facial emotion recognition
	ANUSStressDB	[66]	2014	35 individuals	Thermal and visible spectrum videos	Stress classification (stressed/non-stressed)
	ASCERTAIN	[59]	2016	58 individuals	EEG, ECG, GSR	Emotion and personality recognition
Transportation and Mobility	DeepBreath	[58]	2017	8 individuals	Video Data	Stress recognition
	NNIME	[67]	2017	102 interaction sessions	Audio, video, ECG	Emotion recognition (anger, sorrow, happiness, frustration, etc.)
	KMU-FED	[64]	2018	55 picture sequences	Images of facial expressions	Real-time facial emotion recognition in driving
	CLAS	[57]	2019	62 participants	PPG, ECG, EDA	Cognitive load, affect, stress recognition
	TILES-2018	[68]	2020	212 hospital personnel	ECG, respiratory patterns, speech features	Stress and health monitoring in work environments
	SRAD	[73]	2005	17 recordings	ECG, EMG, GSR, breathing	Stress detection during driving
	State Farm Distracted Driver	[69]	2016	22,000+ labeled images	Image data	Distracted driving detection
	SEED-VIG	[71]	2017	Multiple sessions	EEG	Vigilance, fatigue detection
	MDVFDD	[72]	2017	68 participants	Perinasal EDA, palm EDA, HR, RR	Distracted driving detection
	HDD	[76]	2018	100+ hours of video	Video, GPS, IMU, CAN bus	Driver behavior understanding
Healthcare	Drive&Act	[70]	2019	13 hours of recordings	RGB, depth, thermal, IMUs, 3D skeleton data	In-car activity recognition
	DMD	[75]	2020	41 hours of data	RGB, depth, infrared, eye tracking	Driver distraction, attention analysis
	ADARP	[74]	2022	11 participants	HR, EDA, temperature, movement	Stress and alcohol relapse monitoring
	MPDB	[77]	2024	35 participants	EEG, ECG, EMG, GSR, eye movement	Driving behavior analysis
	CHFDB	[78]	2000	15 patients	ECG	HR variability, arrhythmia detection
	Apnea-ECG	[85]	2000	70 participants	ECG	Sleep apnea detection
	MIT-BIH	[79]	2001	48 half-hour segments	ECG	Arrhythmia detection
	CHB-MIT EEG	[80]	2010	22 patients	EEG	Seizure detection in juvenile epilepsy
	Abdominal ECG	[81]	2012	5 pregnant women	ECG	Fetal and maternal HR monitoring
	BCI IV 2a	[83]	2012	9 participants	EEG	Brain-computer interfaces
Education and Learning	PTB Diagnostic ECG	[84]	2018	549 ECG recordings	12-lead ECG	Cardiac disease detection
	MIMIC-IV	[82]	2020	40,000+ patients	Clinical data, vital signs	ICU patient monitoring
	COPD Electronic Nose	[86]	2021	×	Exhaled breath samples, electronic nose	COPD detection via breath analysis
	StudentLife	[87]	2014	48 students	Mobile phone sensing data	Mental health, academic performance analysis
	xAPI-Educational Mining	[91]	2015	×	Web-based learning data	Educational data mining
	Automated Online Exam Proctoring	[89]	2017	×	Webcam, wearcam, microphones	Online cheating detection
	Student Performance	[90]	2021	326 students	Academic records, Moodle data	Student performance analysis
	ATL-BP	[88]	2022	2,749 video samples	Webcam recordings	Student engagement analysis
	SCB	[92]	2023	4,003 images	Images data	Student behavior detection
	S-Test	[94]	2017	×	physiological signals	Stress level monitoring
Workplace Productivity and Well-being	HR Employee Attrition	[97]	2017	×	HR analytics data	Employee retention analysis
	WESAD	[93]	2018	15 participants	ECG, GSR, EMG, BVP, temperature, acceleration	Stress and affect detection
	DS-3	[95]	2019	×	HR, EDA, voice recordings	Stress reaction analysis
	Employee's Performance	[96]	2023	×	HR analytics data	Employee performance analysis
	Online Retail	[98]	2011	×	Transactional data	Consumer behavior analysis
Marketing and Consumer Behavior	Bank Marketing	[100]	2019	41,188 conversations	Client phone records	Term deposit marketing
	Consumer Complaint	[102]	2019	×	Complaint records	Financial product complaint classification
	UK Consumer Trends	[101]	2023	×	Consumer expenditure data	Economic growth analysis
	Instacart Market Basket	[99]	2024	3 million orders	Online food orders	Market basket analysis

A: Accelerometer, G: Gyroscope, M: Magnetometer, ECG: Electrocardiogram, EMG: Electromyogram, GSR: Galvanic Skin Response, EDA: Electrodermal Activity, PPG: photoplethysmography, HR: Heart Rate, IMUs: inertial motion units, GPS: Global Positioning System, CAN bus: Controller Area Network bus, BVP: Blood Volume Pulse, RR: Respiratory Rate, RGB: Red, Green, Blue, ×: Not Specified.



FIGURE 4. Human behavior analysis applications.

and IoT-edge fusion approaches using hybrid classifiers, integrating IoT-edge data and LR, achieved near-perfect accuracy in remote health monitoring [111]. Similarly, HMM were applied to detect anomalous behavior in remote patient monitoring systems [112], and FPGA-based biosensors were developed for early virus detection [113]. Cognitive impairment detection in healthcare settings has also been explored using SVM, KNN, NN, NB, and DT, with DT achieving 100% accuracy, though the validation was limited to a smaller test size and controlled environment [114].

For mental health, XGBoost and DNN identified biomarkers related to mental disorders connected with air pollution [115]. Voice-based models using DT, CHAID, and CRT were used for depression screening [116]. Additionally, multi-class LR was applied to identify predictors of depression and anxiety during the COVID-19 pandemic [117]. In other health monitoring systems, system dynamics modeling was used for respiratory monitoring via smartphones [118], and telemonitoring systems were employed to manage glucose levels in diabetes patients [119].

Density-based clustering was used for anomaly detection in elderly care, improving behavioral monitoring systems [120].

Table 5 encapsulates the predominant research endeavors within the realm of HBA-based healthcare, providing a succinct overview of each investigation, the resultant performance metrics, and the identified constraints associated with each study.

1) DISCUSSION AND CRITICAL INSIGHTS

The application of HBA in healthcare and mental health demonstrates significant promise, particularly through the use of deep learning and hybrid machine learning models for diagnostics, monitoring, and early detection. LSTM and ensemble methods have proven effective in recognizing physical activities and predicting disease progression, while IoT-based systems enable real-time remote monitoring. However, several limitations persist across the reviewed studies.

A key concern is the limited generalizability of many models due to small, homogeneous datasets or controlled

TABLE 5. Summary of research works on healthcare and mental health.

Ref.	Contribution	Algorithms	Performance	Limitation
[103]	Prediction of intensive care unit (ICU) admission and mortality for patients	Multivariable LR Analysis	AUC for ICU admission: 0.65, AUC for in-hospital mortality: 0.66	Performs poorly for ICU mortality prediction and has limited generalizability
[105]	Diagnosis of chronic kidney disease	SVM, ANN, RF, XGBoost, KNN	Acc: 100% (ANN)	Absence of external evaluation and applicability across various healthcare environments
[108]	IoT-based framework for real-time health and fitness monitoring	LSTM	Acc: 97% for health zone identification, Acc: 89% for gym activity recognition	Limited to a test in a controlled environment
[107]	Predict clinical outcomes for various health conditions	RF, SVM, LR	Acc: 85%	Requires high-quality data from electronic health records
[104]	Early warning system for health crises	GBM, LR	Acc: 80-90%	The study is retrospective and conducted at a single institution, which restricts its generalizability
[110]	Real-time Parkinson's Disease detection using body sensor	VAER and K-Means	Acc: 75%	Struggles with very large datasets and may produce lossy compression leading to some data loss
[111]	IoT-edge fusion approach for remote health monitoring	Hybrid classifier integrating IoT-edge and LR.	Acc: 99.7%	Risk of data breaches stemming from wireless connectivity and constrained edge computer resources
[106]	Hybrid survival assessment model for advanced heart failure patients	BP NN, KNN	Acc: 93.75% (BP NN)	Limited sample size and insufficient data comprehensiveness
[109]	Improving Alzheimer's disease classification.	Ensemble learning combining MFA, MMD, MRD features	Acc: 90%	Significant complexity arises from the ensemble method and necessitates balanced datasets for precision.
[114]	Cognitive impairment detection in healthcare settings	SVM, KNN, NN, NB, DT	Acc: 100% (DT)	Limited validation, smaller test size and controlled environment
[120]	Detecting behavior anomalies for elderly care	Density-Based Clustering, Hierarchical Density Estimates	Acc: 93%	Limited to geriatric care and passive monitoring, it may not extrapolate effectively to other demographics.
[115]	Discovered prospective biomarkers for mental illnesses associated with air pollution	XGBoost, DNN	AUC ROC of 0.792	Data limitations due to uncoded suicide attempts; limited sample size impacting generalizability.
[118]	Smartphone-based systems for respiratory monitoring	System Dynamics Modeling, Breathing Signal Analysis	90% in breathing signal analysis	Demands extensive longitudinal data acquisition and verification, with challenges in practical implementation.
[119]	Telemonitoring for managing glycemic control in Type 2 Diabetes patients	×	The sustainability level rose significantly to 0.87	Restricted examination of telemonitoring effects also comprehensive consequences require investigation.
[116]	Voice-based model for screening depression	DT, CHAID, CRT	Acc: 83.4% (CRT)	Concentrate exclusively on depression; lacks the capacity for real-time adaptation to more intricate mental health conditions.
[117]	Identify predictors of depression and anxiety symptoms in Brazilian adults during COVID-19	Multi-Class Logistic Regression Analysis	Prevalence of depressive symptoms: 70.3%, anxiety: 67.2%	Cross-sectional study constrained by a restricted sample size and a non-representative population resulting from internet Access limitations.
[112]	Detecting anomalous behavior in Remote Patient Monitoring systems	HMM, Baum-Welch Algorithm, Forward-Backward Algorithm	Acc: 98%	Constrained by dataset Dims and specificity to domestic monitoring contexts
[113]	Biosensor for early virus detection in healthcare	×	The impedance measurements aligned with the modeling findings, confirming the sensor's efficacy	Limited by the prototype phase and absence of empirical validation

LR: Logistic regression, LSTM: Long Short-Term Memory, SVM: Support Vector Machine, GBM: Gradient Boosting Machine, IoT: Internet of Things, NN: Neural Network, DNN: Deep Neural Network, DT: Decision Tree, HMM: Hidden Markov Model, MFA: Mean Fractional Anisotropy, MMD: Mean Mean Diffusivity, MRD: Mean Radial Diffusivity, VAER: Variational AutoEncoder Regularization, ANN: Artificial Neural Network, KNN: K-Nearest Neighbors, NB: Naive Bayes, CHAID: Chi-squared Automatic Interaction Detector, CRT: Classification and Regression Tree, RF: Random Forest, ×: Not specified.

testing environments. For instance, while some classifiers achieved near-perfect accuracy, their performance may not hold in diverse, real-world settings. The inconsistency in evaluation metrics and lack of external validation across studies further complicates direct comparisons. In addition, there is a notable absence of longitudinal analyses that could capture behavioral changes over time, which is critical for chronic and mental health conditions.

From a practical standpoint, real-world deployment remains a challenge. Privacy risks, data security in IoT systems, and the interpretability of complex models are often under-addressed. Particularly in mental health applications, models tend to focus on detection without integrating feedback mechanisms or considering socio-environmental variables.

Future research should focus on developing standardized benchmarks, improving model transparency, and validating algorithms in real-world clinical environments. Ethical considerations, including bias mitigation and patient consent in behavioral monitoring, must be embedded into system design. There is also a growing need for multimodal,

explainable, and adaptive systems that can operate reliably in both hospital and home-care settings.

B. MARKETING AND CONSUMER BEHAVIOR

Various ML and DL techniques have been applied to predict consumer behavior and optimize marketing strategies by analyzing multiple data sources such as social media sentiment, reviews, and behavioral patterns. Several studies employed RF models to predict consumer behavior based on environmental, organizational, and interpersonal factors [121], and Sentiment Analysis combined with ML classifiers was used to predict preferences for secondhand luxury products [122]. In e-commerce, ML models were applied to analyze customer behavior, clustering customers and predicting purchase trends for personalized marketing strategies [123], while DL frameworks combining CNN and LSTM networks were used to predict consumer purchase intentions [124].

Studies focusing on repurchase behavior and customer retention used hybrid models, combining CNN with sentiment analysis [125], and models based on the Recency,

TABLE 6. Summary of research work on marketing and consumer behavior.

Ref.	Contribution	Algorithms	Performance	Limitations
[121]	Time-evolving RF classifier to predict consumer behavior based on environmental, organizational, individual, and interpersonal factors	RF	Acc: 90%	Limited feature exploration, focused on specific behavioral factors
[124]	DL framework to predict consumer purchase intentions for online pharmacy products, incorporating trust, ease of use, and social influence factors	CNN, LSTM	Acc: 92%	Limited dataset, focused on specific industry (pharmaceutical)
[122]	ML-based approach for predicting preferences for secondhand luxury products post-pandemic in Indonesia using sentiment analysis	Sentiment analysis, ML classifiers	Acc: 88%	Focus on Indonesian market limits generalizability
[123]	ML model for analyzing customer behavior on e-commerce platforms, focusing on purchase trends and personalization	Clustering, classification, predictive modeling	Acc: 85%	Limited exploration of feature engineering, focused on specific behaviors
[137]	EEG-based consumer choice prediction using spatial attention-enhanced deep transfer learning and 2D CNN for e-commerce products	2D CNN, deep transfer learning	Acc: 95.60%	Limited by small dataset size, niche application
[125]	Deep hybrid learning model to predict customer repurchase behavior for smartphones using user reviews and images	Hybrid model (CNN + sentiment analysis)	Acc: 90%	Focus on smartphones limits broader applicability
[127]	ML model predicting customer return visits to airlines using sentiment analysis from feedback comments and satisfaction ratings	NB, SVM, DT	Acc: 83.42%	Industry-specific, limited to airline service context
[126]	RFM-based customer classification and segmentation model to predict repurchase behavior, using various ML techniques	MLP, SVM, DT	Acc: 97%	Limited to transaction-based data, lacks broader behavioral factors
[133]	Investigated the influence of online search volumes on movie ticket sales, finding a correlation between search volume and revenue	Regression analysis	Acc: 88%	Focused solely on movie industry, lacks general applicability
[136]	Identified green initiatives, AI integration, and experiences as key factors influencing consumer shopping center choices	Text mining, ML classifiers	Acc: 89%	Focused on shopping center reviews, limited generalizability
[134]	Impact of emotional robots on consumer behavior, focusing on emotional expressions and positive reactions	DL	Acc: 90%	Limited to the study of emotional robots
[135]	Investigated the role of group recommender systems in improving decision-making for hedonic products	Group recommender systems	Acc: 85%	Focused on group-based decision-making, niche application
[129]	Developed a NB classifier to classify hospitality service reviews into positive/negative categories	NB	Acc: 85%	Limited to hospitality industry reviews
[132]	Price Elasticity Impact Learning (PEIL) model for optimizing promotions in e-commerce without historical price elasticity data	GBM, log-log demand model	Acc: 91%	Limited to consistent pricing strategies
[128]	Developed a model to estimate classifier certainty in predicting customer churn in telecommunications using distance factors	NB, clustering	Acc: 83%	Limited to telecommunications industry, lacks broader context
[130]	Developed ML models to predict the helpfulness of online consumer reviews based on textual features	GB, ensemble learning	Acc: 87%	Focused on textual data, lacks integration with other consumer behavior aspects
[131]	Data mining approach to identify incentivized online reviews based on review length, helpfulness, and sentiment scores	DT (C5.0, C&RT), RF, VADER algorithm	Acc: 75%	Focused on Amazon data, limited to incentivized reviews context

RF: Random Forest, LSTM: Long Short-Term Memory, NB: Naïve Bayes, SVM: Support Vector Machine, DT: Decision Tree, MLP: multilayer perception, DL: Deep Learning, GBM: Gradient Boosting Machines, ML: Machine Learning, GB: Gradient Boosting, CNN: Convolution Neural Network.

Frequency, and Monetary (RFM) framework, combined with various ML techniques such as Multilayer Perceptrons and SVM [126]. Additionally, Naïve Bayes and Decision Trees models were employed for predicting customer return visits based on sentiment analysis from feedback [127] and for customer churn prediction in the telecommunications industry [128].

For analyzing consumer reviews, ML models such as Naïve Bayes were applied to classify hospitality reviews [129], while ensemble learning techniques were used to predict the helpfulness of online reviews [130]. Sentiment analysis combined with data mining techniques was also applied to detect incentivized reviews [131]. Other studies explored the impact of Price Elasticity Impact Learning (PEIL) models to optimize product promotions [132] and used regression analysis to investigate the influence of online search volumes on movie ticket sales [133].

Emotional factors have also been explored, with studies using DL models to assess how emotional robots influence consumer reactions [134] and group recommender systems to improve decision-making for hedonic products [135]. Studies examining sustainable consumer behavior identified green initiatives, AI integration, and memorable experiences as key factors influencing shopping choices using text mining and ML classifiers [136]. Additionally, EEG-based

consumer choice prediction used spatial attention-enhanced deep transfer learning and 2D CNN for e-commerce products, achieving an accuracy of 95.60%, though limited by small dataset size and niche application [137].

Table 6 provides a comprehensive summary of recent research in marketing and consumer behavior.

1) DISCUSSION AND CRITICAL INSIGHTS

The application of HBA in marketing and consumer behavior has evolved significantly through the integration of machine learning, deep learning, and sentiment analysis techniques. Many studies have shown promising results in predicting purchase intent, customer retention, and review helpfulness by leveraging textual, behavioral, and biometric data. Random Forests and CNN-LSTM hybrids are particularly popular due to their robustness in handling high-dimensional and sequential data, while Naïve Bayes and decision trees continue to serve well in lightweight classification tasks.

However, a common limitation across studies is the narrow focus of datasets—either tied to specific industries (e.g., pharmaceuticals, airlines) or regions (e.g., Indonesia)—which restricts the generalizability of the findings. Additionally, while accuracy metrics are often high, there is limited discussion on the interpretability of these models, especially

when used in customer-facing applications. The absence of standardized behavioral features and inconsistent use of evaluation metrics also hinders cross-study comparison.

Another gap lies in the underutilization of multimodal data fusion. Although some studies explore EEG signals or image data, these are typically isolated cases and not yet standard practice. The emotional and psychological dimensions of consumer decision-making—while crucial—remain only partially addressed, often through sentiment labels rather than deeper cognitive or behavioral modeling.

Future research should explore the integration of richer data types (e.g., gaze tracking, voice tone, physiological signals) for a more holistic understanding of consumer behavior. There is also a need for developing interpretable and privacy-preserving models that maintain fairness across demographics. Research should additionally aim at real-time systems for adaptive marketing and customer feedback loops, as well as benchmarking protocols that allow replicable and comparative evaluation across studies.

C. EDUCATION AND LEARNING

In the field of education, various ML and DL techniques have enhanced both teaching and learning experiences [138]. Intelligent Tutoring Systems (ITS), which leverage NLP, Reinforcement Learning (RL), and adaptive learning models, have been effective in subjects such as mathematics, programming, and chemistry, providing personalized learning experiences and real-time feedback to improve student performance [140], [141], [142], [143], [144], [145]. These systems have been shown to improve learning outcomes across different domains.

Early Warning Systems (EWS) [139], [146] for dropout prediction use models such as RF, LR, SVM, and DL to analyze student performance and engagement data to identify students at risk of dropping out [147], [148], [149], [150], [151]. These systems help institutions intervene early and improve student retention rates.

Classroom Engagement Monitoring (CEM) [152] is another application where AI-driven models, including SVM, DT, KNN, and CNN, have been used to monitor student participation in real-time, particularly in remote learning environments. These systems assess emotional, behavioral, and cognitive engagement, offering insights into student performance and helping educators create more effective learning environments [153], [154], [155], [156].

Automated Attendance Systems (AAS) have improved the accuracy and efficiency of attendance tracking through facial recognition and CNN models such as AlexNet, GoogleNet, and SqueezeNet [157], [158], [159]. These systems streamline attendance management by using biometric data, reducing manual errors and saving time in large classrooms.

Automated Essay Scoring (AES) has transformed the grading process through models such as BERT, Recurrent Neural Networks (RNN), and Generative Adversarial Networks.

These algorithms offer a consistent and unbiased approach to essay evaluation, with systems such as EASE and topic-aware BERT providing high accuracy and reliability [160], [161], [162], [163]. Additionally, active learning methods have been explored to improve the performance of AES systems [164].

Pronunciation Training Tools (PTTs), powered by Automatic Speech Recognition (ASR), have been employed in language learning to improve phonetic accuracy. Models such as GMM-HMM, CRDNN, and wav2vec2 have demonstrated success in enhancing pronunciation skills for non-native speakers [165], [166], [167].

Table 7 summarizes research work on the education and learning field.

1) DISCUSSION AND CRITICAL INSIGHTS

The application of HBA in education has seen rapid advancement through AI-driven methods that enhance personalization, engagement, assessment, and early intervention. ITS and EWS have demonstrated measurable impacts on learning outcomes and student retention, while classroom engagement monitoring and automated assessment tools have addressed longstanding challenges in scalability and feedback delivery. These systems commonly employ CNNs, RF, SVM, and more recently, advanced models like BERT and GANs, reflecting a trend toward deep contextual understanding and automation.

Despite these advancements, several limitations are evident across the reviewed studies. Many models are tested in controlled or small-scale environments, raising concerns about their real-world generalizability, particularly across diverse educational contexts or cultural settings. There is also a notable imbalance between cognitive and non-cognitive behavior analysis; while performance prediction is well-studied, emotional and social dimensions of learning behavior remain underexplored. Furthermore, interpretability is often sacrificed for performance—especially with deep models like BERT—hindering educator trust and adoption in practice.

Another critical gap is the lack of long-term, longitudinal studies that assess the impact of these systems over time. For example, while dropout prediction models report high accuracy, few explore the effectiveness of subsequent interventions. Similarly, automated essay scoring and pronunciation tools offer impressive metrics but rarely account for student learning curves or pedagogical feedback integration.

Future research should prioritize interpretable AI models that offer transparent feedback to both educators and learners. There is also a need for systems that integrate multimodal data (e.g., visual, auditory, behavioral) to capture richer insights into student engagement. Longitudinal evaluation frameworks, privacy-aware architectures, and culturally inclusive datasets will be essential to scale AI-based education systems responsibly and equitably across diverse learning environments.

TABLE 7. Summary of research work on education and learning.

Application	Ref.	Algorithms	Performance
ITS	[145]	NLP and RL	Learning gain: 48.53%
	[140]	Optimized Ant Colony, Case-Based Reasoning	Proposed ITS enhance programming performance and offer an immersive experience.
	[141]	Adaptive learning models	Beta coefficient of 0.02 ($t = 25.59$, $p < .001$)
	[142]	EMT, BM, LSA, AAA	Pre-test score: 64.20, Post-test score: 83.41, with a significant improvement ($t = -8.418$, $p < .001$), effect size = 1.091
	[143]	SEM	The model had strong predictive capability, accounting for 69.6% of the overall variation.
	[144]	Adaptive Stepped Support tools, Algorithm for hints	Resolved an average of 85% of the difficulties during problem-solving tasks in organic chemistry.
EWSDP	[147]	Logistic Model, Additive Logistic Model, SVM, RF	MAE: 17.6%
	[148]	CNN, RF, LR, Cox Proportional Hazards, DeepSurv	Acc: 0.855 - 0.987, AUC: 0.909 - 0.963, C-index: 0.89 - 0.95
	[149]	DT	C-index: 0.7, AUC: 0.86
	[150]	LR	AUC: 0.70 - 0.82, Lift: 1.68 - 3.88
	[151]	SVM, RF, LR, NB, Sequence Mining Techniques (cSPADE, PrefixSpan, BIDE, iBCM)	AUC: 0.66 - 0.9, Lift: 2.23 - 2.43
CEM	[155]	LBP, PnP, HOG, SVM, KNN, RF, DT	Acc: 78.6%(RF)
	[156]	ENQUA model	Masterclass engagement: 33%, Hands-on engagement: 66%
	[153]	Dlib Model, CNN, SVM	Acc: 67.7%
	[154]	Metacognitive engagement	Acc: 92.2%
AAS	[157]	CNN using AlexNet, GoogleNet, SqueezeNet	Acc: 95.02%
	[159]	SVM, NN, CNN	99.33% (Tuned SVC), 98.74% (Linear SVC)
	[158]	LR, LDA, KNN	97.29% (LDA), 93.22% (LR)
AES	[160]	EASE, DNN, variants of RNN, BERT model.	Acc: 94%
	[161]	Multiple AES models: EASE (SVR), EASE (BLRR), XGBoost, RNN, SkipFlow, Hybrid-BERT.	QWK average: 75.62%
	[162]	Topic-aware BERT	QWK: 78.9%
	[163]	BERT, GAN	QWK improvement up to 86.5%
	[164]	Active Learning Methods	95% QWK with 1.8% of data
PTTs	[165]	ASR	Success rate for Constant players: 68.3% (production) and 97% (discrimination)
	[166]	ASR	Pre-test mean score: 0.85, Post-test mean score: 2.59, Improvement mean: 1.74 ($Z = -13.551$, $p < 0.001$), with a large effect size ($r = 0.50$).
	[167]	GMM-HMM, CRDNN, DT for ASR	UAR score: 91.47% on SweSSD

ITS: Intelligent Tutoring Systems, ML: Machine Learning, NLP: Natural language processing, RL: Reinforcement Learning, SVM: Support Vector Machine, RF: Random Forest, CNN: Convolutional Neural Network, LR: Logistic regression, cSPADE: Constraint-based Sequential Pattern Discovery using Equivalence classes, PrefixSpan: Prefix-projected Sequential pattern mining, BIDE: Bidirectional Extension-based Sequential pattern mining, iBCM: Instance Based Constraint Mining, LBP: Local Binary Patterns, PnP: Perspective-n-Point, ECG: Electrocardiogram, SEM: Structural Equation Modeling, NN: Neural Network, LDA: Linear Discriminant Analysis, KNN: K-nearest neighbors, QWK: quadratic-weighted Kappa, GAN: Generative adversarial network, IRT: Item-Response-Theory, EWSDP: Early Warning Systems for Dropout Prediction, BERT: Bidirectional encoder representations from transformers, AL: Active Learning, ASR: Automatic Speech Recognition, GMM: Gaussian Mixture Model, HMM: Hidden Markov Model, CRDNN: Convolutional Recurrent Deep Neural Network, UAR: Unweighted Average Recall, CEM: Classroom Engagement Monitoring, AAS: Automated Attendance Systems, AES: Automated Essay Scoring, PTTs: Pronunciation Training Tools, NB: Naive Bayes, DT: decision tree, HOG: histogram of oriented gradient, ENQUA: ENgagement QUAntification, EASE: Enhanced AI Scoring Engine, DNN: Deep neural network, BLRR: Bayesian linear ridge regression, SVR: Support Vector Regression, EMT: Expectation and Misconception Tailored Dialogue, BM: Block Matching, LSA: Latent Semantic Analysis, AAA: Automated Analysis Approach.

D. WORKPLACE PRODUCTIVITY AND EMPLOYEE WELL-BEING

In the area of workplace productivity and employee well-being, ML techniques have been widely applied to assess mental health, predict employee attrition, and optimize workplace environments. Various studies focused on mental health analysis, using models such as DT and ANN to predict mental health issues based on factors such as stress, anxiety, and work-life balance [168], [169], [170]. DL methods combined with fuzzy clustering have also been employed to monitor daily stress and mood patterns [171].

Employee attrition prediction has been another key area of research, where models such as RF, AdaBoost, and XGBoost were used to analyze HR data and predict employee turnover [172], [173]. Other studies applied KNN, LR, and MLP to predict churn and identify factors influencing employees' decisions to leave [174], [175].

Several studies explored the physical and emotional work environment, using DT to correlate environmental factors such as lighting and temperature with employee productivity [176]. Mood recognition systems integrating KNN, DT, and wearable sensors were also used to monitor employee well-being and productivity in real-time [177]. CatBoost

TABLE 8. Summary of research work on workplace productivity and employee well-being.

Ref.	Contribution	Algorithms	Performance	Limitations
[168]	Mental health analysis in workplace using ML	LR, KNN, DT, RF	Acc: 81% (DT)	Limited to tech employees, needs generalization, lacks wearable device integration
[185]	Applicant prediction model based on employee perception	NB, Association Rule Mining	×	Limited to a small group of employees, focused on one job position, needs validation on larger datasets
[181]	Anxiety detection and management app for IT employees	Rule-based ML	×	Limited to Sri Lankan IT industry, lacks validation beyond local context, limited work-specific activities
[184]	Analysis of AI's impact on employee productivity using BYOD strategy	RF, DT, LASSO, Linear Regression	×	Limited sample size, lacks generalization across industries, no long-term validation
[172]	Employee attrition prediction using ML	RF, AdaBoost	Acc: 87% (AdaBoost)	Limited features considered, lacks other critical factors such as workplace culture
[171]	Daily stress and mood recognition system using DL	SVM, KNN, DT, Fuzzy Clustering	Acc: 91% (Personalized SVM), 83% (Generic SVM)	Limited to a small sample size, no real-time implementation
[176]	Smart indoor workplace system correlating physical environment with work efficiency	DT	0.89 AUC (Air conditioning), 0.50 AUC (Lighting)	Limited sample size, random prediction for lighting, lacks diverse environmental conditions
[174]	Employee churn prediction using KNN	KNN, NB, LR, MLP	Acc: 93% (KNN)	Lacks real-world scenario testing
[177]	Mood recognition system using smartphones and wearable sensors	KNN, DT, Bagged Ensembles (BE-DT)	Acc: 70.6% (Bagged Ensembles)	Small-scale study, limited to office environment, lacks long-term validation
[175]	Employee attrition prediction using LR	LR	Acc: 90.14%	No comparison with other models, lacks external validation
[169]	Stress detection in IT employees using ML	RF, NB, DT, SVM, KNN	Acc: 85.07% (RF)	Limited to IT sector, lacks real-time application, small sample size
[178]	Employee satisfaction optimization using health and safety measures	CatBoost, PCA, SHAP analysis	×	Focused on manufacturing sector, lacks generalization across industries, requires further real-world validation
[173]	Employee attrition prediction using XGBoost	XGBoost	Acc: 89% (XGBoost)	Focused on HR dataset, lacks external validation, no real-world deployment
[170]	Predicting employee mental health using NN	ANN, CNN, RNN	Acc: 92.2% (ANN), 89.1% (CNN), 86.9% (RNN)	Focused on mental health dataset, no real-world deployment, lacks cross-validation
[182]	Work and job environment prediction using transfer learning	KNN-SVM, SVM, KNN	Acc: 96.58% (KNN-SVM), 94.24% (SVM), 91.35% (KNN)	Focused on service industries, lacks generalization to other sectors, no real-time deployment
[179]	Real-time worker stress prediction in smart factories using IoT and ML	XGBoost, LR, DT, RF, KNN, SVM	Acc: 99% (XGBoost)	Focused on assembly lines, no real-world deployment, limited to physiological data
[180]	Real-time employee emotion detection using ML	AdaBoost	×	Focused on facial expressions, limited to six emotions, no real-world validation
[183]	Stress and productivity performance modeled using binary decision automata	Binary Decision Automata	×	Focused on stress modeling, lacks real-world data, simplified agent behaviors

KNN: K-Nearest Neighbors, DT: Decision Tree, PCA: Principal Component Analysis, ANN: Artificial Neural Network, NB: Naive Bayes, RF: Random Forest, ML: Machine Learning, RNN: Recurrent Neural Network, KNN: K-Nearest Neighbors, SVM: Support Vector Machine, MLP: Multi-layer Perceptrons, LASSO: Least Absolute Shrinkage and Selection Operator, Acc: Accuracy, LR: Logistic Regression, ×: Not Specified, CNN: Convolution Neural Network.

and Principal Component Analysis (PCA) were combined with SHAP analysis to optimize employee satisfaction in workplace settings [178].

Real-time stress and emotion detection systems have also gained attention, with models such as XGBoost, LR, and SVM being used in factory environments to monitor worker stress and provide real-time feedback [179], [180]. Sentiment and emotional analysis have been applied to manage workplace anxiety and improve performance, particularly in the IT and manufacturing sectors [181].

In improving overall workplace productivity, studies have employed predictive models to analyze job environments and employee satisfaction. Transfer learning techniques were used to predict work and job factors that influence employee satisfaction in the service industry [182]. Agent-based models were applied to simulate the impact of stress and productivity [183], while one study examined the Bring Your Own Device (BYOD) strategy and its positive effects on employee productivity and satisfaction through enhanced transparency and workflow management [184].

Lastly, applicant prediction models based on employee retention and performance factors were developed using NB

and association rule mining, offering insights into better hiring practices to reduce employee turnover [185].

Table 8 summarizes research work related to workplace productivity and employee well-being, highlighting their contributions, algorithms used, performance metrics, and limitations.

1) DISCUSSION AND CRITICAL INSIGHTS

Research in workplace productivity and employee well-being has embraced a diverse range of machine learning techniques to address mental health, attrition, satisfaction, and performance optimization. Predictive models such as Random Forests, AdaBoost, and neural networks have demonstrated high accuracy in identifying at-risk employees and forecasting churn, while real-time systems using wearable sensors and IoT data offer new possibilities for continuous monitoring of stress and mood in the workplace.

However, a closer examination reveals several recurring limitations. Many studies are based on narrow datasets restricted to specific sectors (e.g., IT, manufacturing) or geographic regions, limiting the generalizability of their findings. Furthermore, while high classification accuracy

is frequently reported, few models have been externally validated or tested in live organizational settings, which raises questions about their practical deployment. There is also a lack of standardization in feature sets and outcome definitions, particularly in studies focused on emotional well-being, which hampers cross-study comparison.

In addition, while some models incorporate environmental or behavioral data, most are heavily reliant on survey-based or HR datasets, which can be biased or outdated. Interpretability remains a challenge in complex models like ensemble learners and deep neural networks, making them less suitable for sensitive domains like mental health where actionable insights are crucial. Very few studies account for longitudinal dynamics or explore how behavioral patterns evolve over time, and almost none integrate ethical frameworks to address privacy concerns in workplace surveillance.

Future research should aim for more holistic modeling by combining behavioral, physiological, and environmental data using multimodal and explainable AI frameworks. Longitudinal studies that assess the sustained impact of interventions or predictive systems are urgently needed. Moreover, building adaptable systems that can generalize across different industries while preserving user privacy will be essential for real-world adoption and ethical integration of AI in workplace management.

E. CRIMINAL JUSTICE AND PUBLIC SAFETY

In criminal justice and public safety, a wide range of ML and DL techniques have been applied to improve traffic violation detection, crime prediction, vehicle tracking, and criminal recognition.

For traffic violation detection, various DL models have been used, such as Faster R-CNN and ShuffleNet v2 for red-light violation detection [186], and YOLOv4, YOLOv5, and YOLOv8 for detecting helmet violations and other traffic infractions [187], [188], [189]. Additionally, automatic detection systems using radar and video-based methods have reduced accidents and fatalities [190]. Mask R-CNN, DenseNet-121, and ResNet-101 were employed to detect helmets and number plates with high accuracy [191]. Vehicle tracking and traffic surveillance models, such as YOLO, combined with tracking algorithms such as the Kalman filter and Hungarian algorithm, have demonstrated high precision in vehicle tracking [192]. YOLOv5 combined with strongSORT has also shown efficiency in detecting traffic violations using dashcam footage [193].

In the domain of crime prediction and analysis, a variety of models have been employed. LSTM, XGBoost, and ARIMA have been applied in cities such as Chicago and Los Angeles, achieving high accuracy in predicting crimes [194]. SARIMA, LSTM, and CHD models have been tested to improve prediction accuracy [195]. More advanced DL models such as DeepCrime, Multi-View Deep Spatial-Temporal Network (MiST), and CrimeForecaster have shown reduced prediction errors [196]. Spatio-Temporal

LSTM, Attention LSTM, and Bi-LSTM models demonstrated high prediction accuracy for crime data in urban areas such as Chicago and San Francisco [197]. LR and correlation analysis models were applied to predict crime based on historical data [198], [199], and LASSO and Geographically Weighted Regression (GWR) models were used for predicting property crimes [200]. SARIMA models have also been applied to forecast crime rates with high accuracy [201].

Crime risk assessment research employed models such as binomial LR and semantic segmentation to assess environmental factors influencing crime [202]. In cryptocurrency crime detection, correspondence analysis and correlation models provided insights into crime risk factors [203]. Thresholded ridge estimation models were used to detect crime hotspots in cities such as Chicago [204], and seasonal crime rates were analyzed using Poisson state-space models [205]. Studies investigating crime variables and their relationships used Pearson correlations to analyze data [206].

In criminal detection and tracking, Faster R-CNN, CNN, and RNN models have been applied to recognize individuals in public datasets and track criminals with high accuracy [207], [208], [209]. Heuristic algorithms were used to enhance crime prediction and solve complex problems in large datasets [210]. In addition, SVM, RF, and Naive Bayes (NB) models were applied to detect malicious activities on social media, showcasing the power of ML for crime detection across different platforms [211].

For patrol optimization, techniques such as Genetic Algorithm (GA) and Cuckoo Search (CS) were used to optimize police patrol routes and improve response times [212], [213]. Studies using the Rabbit Walk Algorithm demonstrated improved patrol coverage and safety in smart cities [214]. Additionally, collaboration between UAVs and patrol cars was shown to enhance emergency response efficiency [215]. Multi-agent RL was applied to further optimize response times and improve patrol strategies [213].

Table 9 presents a detailed description of current research in Criminal Justice and Public Safety.

1) DISCUSSION AND CRITICAL INSIGHTS

The integration of machine learning and deep learning in criminal justice and public safety has led to considerable advances in areas such as traffic violation detection, crime prediction, criminal tracking, and patrol optimization. Models like YOLOv5, Faster R-CNN, LSTM, and XGBoost have consistently demonstrated strong performance across detection and forecasting tasks, enabling more accurate monitoring and faster response times. Crime prediction, in particular, has benefited from spatio-temporal modeling techniques (e.g., Bi-LSTM, SARIMA), which improve predictive power by incorporating historical, seasonal, and geographic data patterns.

Despite these technical achievements, several challenges and limitations persist. A significant portion of the reviewed

TABLE 9. Summary of research works on criminal justice and public safety.

Applications	Ref.	Algorithms	Performance
Traffic Violation Detection	[187]	YOLOv5	Precision: 96.31%
	[188]	YOLOv8	mAP: 0.5861
	[186]	Faster R-CNN and ShuffleNet v2	Improvement in Acc: 12%
	[190]	Automatic detection methods, including radar and video-based detection	Reduced Accidents by 22%, deaths by 27%
	[191]	Mask R-CNN, DenseNet-121, ResNet-101, Inception ResNet-V2	Helmet detection: 98%, Number plate detection: 96%
	[193]	YOLOv5, strongSORT	Detection of violations: 98.52%
Vehicle Tracking and Traffic Surveillance	[189]	YOLOv4, Trapezium Bounding Box Regressor, Amodal Regressor	Precision: 84.44% (Triple-Riding Violations), Precision: 99.01% (Helmet Violations)
	[192]	YOLO, Kalman filter, and Hungarian algorithm	Detection: 94.10%, Tracking: 92.50%
	[216]	VGG, YOLOv5, Hashgraph algorithm	Recognition Acc: 89% ,Localization Acc: 92%
Crime Prediction and Analysis	[217]	HAGRU model, MLP, attention mechanism	F-score: 96.71%, Detection Rate: 96.32%
	[198]	LR, Correlation Analysis	Crime Prediction: 61%
	[199]	Correlation Analysis, Gini Coefficient	Precision: 66%
	[218]	Segmented LRs	Best result with an R^2 of 0.77 (Cyber-Dependent Crime)
	[206]	LR	Correlations for theft: 0.63, for assault/muggings: 0.67
	[195]	SARIMA, LSTM, CHD	MAPE: 5.09%-13.37%
	[194]	LSTM, XGBoost, ARIMA	Acc: 94% (XGBoost)
	[200]	LASSO, GWR, regression	RMSE: 2.67, R^2 : 0.59 for property crimes
	[201]	SARIMA	Forecasting Acc within 5-10%, Relative error: 0-9.4%
	[197]	Spatio-Temporal LSTM, Attention LSTM, Bi-LSTM	R^2 : 84% to 97% (Chicago), R^2 : 92% to 97% (San Francisco)
	[219]	RF, XGBoost, DNN	Acc: 85%
	[196]	DC, MiST, CF, HAGEN, ST-SHN, ST-HSL, AIST	MAE: 0.42-0.10, RMSE: 0.35-0.90
	[220]	STL, SPPT	Crime rates fluctuation percentages: Theft: 22.2%-33.7%, Assault: 31%-37%
	[205]	Poisson state-space model, cluster analysis	Explained 95.2% of waveforms similarity
Criminal Detection and Tracking	[207]	TPA, PP, BN	Tracking Acc: 87%
	[210]	Heuristic Algorithm	Acc: 92%
	[211]	SVM, RF, NB, DT	Acc: 97%
	[208]	Faster R-CNN	Acc: 99%
	[209]	CNN, RNN	Acc: 85%
Patrol Optimization	[212]	GA, CS	Efficiency: 90%, Robustness: 85%
	[214]	Rabbit Walk Algorithm	Patrol Coverage: 88%
	[221]	GA, Ant Colony Optimization, Tabu Search, Particle Swarm Optimization	Route Optimization: 84%
	[215]	Air-ground Cooperative Patrol Algorithm	Reduction in response time: 85%
	[204]	Thresholded Ridge Estimation	Acc: 78%
Crime Risk Assessment	[213]	Multi-Agent RL	Response time reduction: 90%
	[203]	Correspondence analysis, correlation models	Chi-square test: $\chi^2=210.47$, p-level < 0.01
	[202]	Binomial LR	AUC: 0.70

YOLO: You Only Look Once, SVM: Support Vector Machine, HAGRU: Hierarchical Attention Gated Recurrent Unit, CNN: Convolution Neural Network, MLP: Multi-layer Perceptrons, DL: Deep Learning, SARIMA: Seasonal AutoRegressive Integrated Moving Average, ARIMA: AutoRegressive Integrated Moving Average, LASSO: Least Absolute Shrinkage and Selection Operator, GWR: Geographically Weighted Regression, BiLSTM: Bidirectional Long Short-Term Memory, MiST : Multi-View Deep Spatial-Temporal Network, GA: Genetic Algorithm, CS: Cuckoo Search, RL: Reinforcement learning, RF: Random Forest, DNN: Deep Neural Network, VGG: Visual Geometry Group, STL: Seasonal-Trend Decomposition using Loess, SPPT: Spatial Point Pattern Test, DT: Decision Tree, MAPE: Mean Absolute Percentage Error, DC: DeepCrime, CF: CrimeForecaster, HAGEN: Homophily-Aware Graph Convolutional Recurrent Network, LR: Logistic Regression, ST-SHN: Spatial-Temporal Sequential Hypergraph Network, NB: Naive Bayes, ST-HSL: Spatial-Temporal Hypergraph Self-Supervised Learning, AIST: Attention-based Interpretable Spatio-Temporal Model, TPA: Tracking Path, PP: Path Prediction, BN: Back Notification, Acc: Accuracy, LSTM: Long Short-Term Memory.

studies rely heavily on datasets from specific urban centers (e.g., Chicago, San Francisco), which limits their generalizability to other regions or rural areas. Moreover, while many models report high precision or accuracy, few are deployed or validated in real-world policing or public safety contexts, which raises concerns about practical viability, ethical implications, and public trust. The lack of standard evaluation benchmarks across studies also complicates model comparison.

Another key concern is the limited integration of social, economic, and demographic variables in most prediction

models. As crime is a socially complex phenomenon, models that overlook these dimensions may reinforce existing biases or produce misleading results. Similarly, in criminal detection and surveillance, the ethical considerations around facial recognition, video analytics, and privacy are often under-discussed, despite their critical importance in public safety applications.

Future research should emphasize building transparent and interpretable models that account for ethical constraints, data bias, and explainability—particularly in high-stakes applications like criminal profiling and patrol deployment.

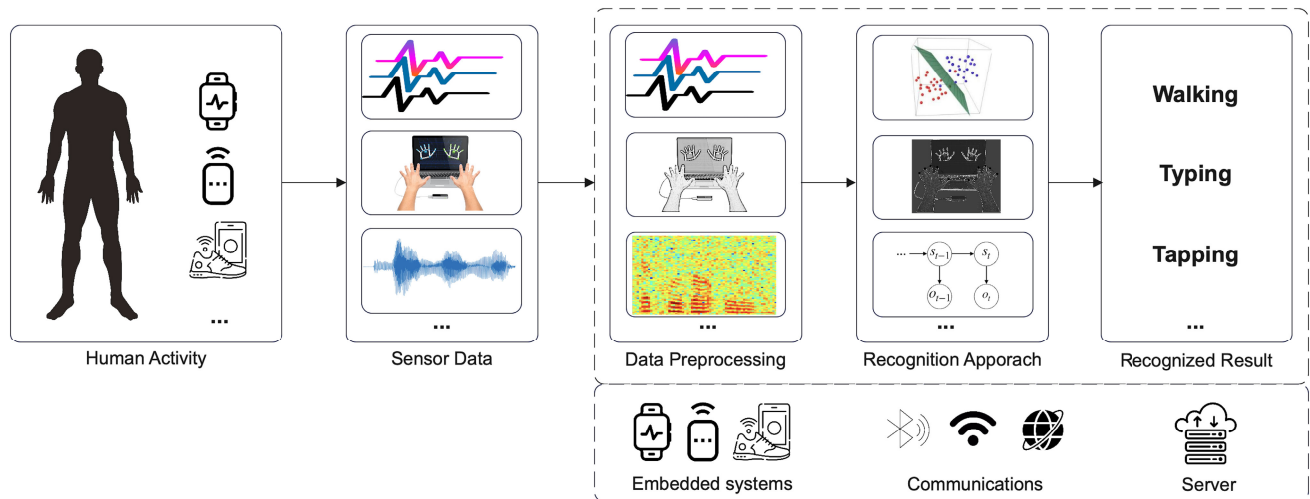


FIGURE 5. The HAR workflow from data collection to activity recognition. The process begins with monitoring human activity using various wearable and embedded sensors, such as smartwatches, smartphones, and IoT devices. These sensors capture raw data, including motion, position, and audio signals, which then undergoes preprocessing to filter noise and extract relevant features. The preprocessed data is fed into recognition models that employ various approaches, such as classification and sequence modeling, to identify specific activities like walking, typing, or tapping. The workflow incorporates embedded systems for data acquisition, communication protocols (e.g., Bluetooth, Wi-Fi) for data transmission, and servers for additional processing and storage.

The development of standardized datasets, domain-specific evaluation metrics, and participatory frameworks involving law enforcement and civil rights organizations will be essential to ensure fairness, effectiveness, and societal acceptance of AI systems in this field.

F. HUMAN ACTIVITY RECOGNITION

HAR plays a pivotal role in understanding and analyzing human behavior across various contexts [7], [8], [9]. By leveraging sensor data and ML algorithms, HAR enables the detection and interpretation of physical activities. Figure 5 depicts HAR's structure. Below are key contributions in this field:

CNN-based models have been widely applied for recognizing daily activities, gestures, and locomotion across multiple studies. These models have proven effective in handling sensor data for activity classification, as seen in works such as [222], [223], [224], [225], [226], [227], [228], [229], [230], and [231], where convolutional networks are used either standalone or in hybrid configurations. Additionally, CNNs have been combined with attention mechanisms and autoencoders to improve performance in both locomotion and gesture recognition tasks [290], as demonstrated in [232], [233], [234], and [235]. Other CNN variants such as HS-CNN have shown strong results, achieving Acc of 97.28%, 93.75%, 99.02%, and 79.02% as reported in [236].

LSTM networks have been another dominant model for HAR, especially in tasks that require capturing long-range temporal dependencies. Studies such as [237], [238], [239], [240], [241], [242], and [243] have leveraged LSTMs for detecting a variety of human activities, including daily

tasks, locomotion, and health-related actions. In several cases, LSTMs were integrated with other techniques, such as GANs [244], attention mechanisms [245], or hybrid models such as CNN-LSTM [246], to further enhance activity recognition across multiple environments.

Generative models such as GANs and autoencoders have also been explored for HAR. Conditional GANs were employed to improve the recognition of activities involving complex motions, as seen in [247], while autoencoders were applied in unsupervised or semi-supervised contexts for feature extraction and dimensionality reduction, as demonstrated in [248], [249], [250], [251], and [252]. These approaches have helped address challenges related to data scarcity and model efficiency.

RNN variants, including structural RNNs and GRU-based networks, have been adopted for group activity recognition and other complex tasks. In particular, studies such as [253], [254], [255], and [256] focused on improving HAR performance by leveraging spatio-temporal attention or integrating multiple recurrent layers for better modeling of sequential data. Additionally, hybrid models combining RNNs with CNNs or attention mechanisms have been explored in works such as [257], [258], and [259], where self-supervised learning combined with multi-task learning led to an Acc of 82.9%.

Finally, transformers and other innovative architectures have emerged in HAR research. Vision Transformers (ViT) were used to improve activity recognition from radar data [260], while other studies have explored the application of transformers in combination with CNNs and LSTMs to enhance the efficiency and Acc of HAR systems [261], [262].

TABLE 10. Summary of research work on HAR.

Activity	Ref.	Algorithms	Performance
Locomotion Activity	[247]	CGAN + VGG-LSTM	84.7%, 76.7% (F1) for respective data
	[230]	CNN	Acc: 89.5%
	[248]	Continuous AE	Acc: 99.3%
	[223]	CNN	97% (F1)
	[225]	AE+CNN	Acc: 95.81%
	[232]	CNN + AE	Acc: 94.2%
	[251]	SAE	Acc: 96.4%
	[238]	LSTM	Acc: 90.5%
	[239]	LSTM	Acc: 95.4%
	[240]	SDAE	94% (F1 score)
	[227]	CNN	Acc: 94.8%
	[228]	CNN	Acc: 92.38%
	[241]	LSTM + Attention Model	Acc: 96.1%, 89.9%, 85.5% (F1)
	[256]	Fuzzy Convolutional Attention-based GRU	Acc: 94.32%
	[257]	CNN-GRU (ICGNet)	Acc: 99.25% (MHEALTH), 97.64% (PAMAP2)
	[260]	LH-ViT	Acc: 99.7%, 92.1%
	[250]	SAE	Acc: 85%
	[233]	CNN + GRU	Acc: 96.5%, 93.1%, 89.3% for respective datasets
Daily Activities	[226]	CNN	88% (F1)
	[234]	1DCNN + BiLSTM + Attention mechanism	Acc: 99.39%
	[222]	CNN	54.7% (F1), 89.6% (Acc)
	[237]	LSTM	Acc: 96.7%, 97.8%, 92%, 93%, 92.6%
	[224]	CNN	Acc: 91.72%, 67.04%
	[253]	Structural RNN	Acc: 89.1%, 89.3%
	[249]	SAE	Acc: 98%, 53.8%, 62.6%, 6.7%
	[229]	GCN+MLP	Acc: 29.4%, 51.6%
	[231]	CNN	Acc: 40.5% (Top-5), 62.7%
	[244]	GAN + LSTM	Acc: 91.7%, 93%
	[243]	LSTM ensembles	Acc: 72.6%, 85.4%, 92.4%
	[254]	Binarized BLSTM	Acc: 76%, 92%
	[262]	2D CNN + 3D Transformer	Acc: 99.7%
	[242]	CNN + LSTM	Acc: 90.89%
	[246]	3D CNN + LSTM	Acc: 95.6%
	[261]	Transformer Model	Acc: 99.2%
	[235]	Heterogeneous CNN	91.55% (F1 Score)
	[236]	HS-CNN	Acc: 97.28%, 93.75%, 99.02%, 79.02%
Gestures	[255]	RCNN-BiGRU + MPA variants	Acc: 94.06%
	[258]	Spatial Attention-Aided CNN with GA	Acc: 98.3%-99.9%
	[259]	Self-supervised learning (ResNet + multi-task learning)	Acc: 82.9%
	[245]	GRU-INC (GRU + Attention Mechanism + Inception + CBAM)	Acc: 96.27%, 90.05%, 99.12%
	[252]	DBM	Acc: 73%, 93%, 94%
	[263]	Localization algorithm	100% recognition rate, 0.06m localization error
	[264]	SVM, MobileNetV2, Transfer Learning	Acc: 98.28%, 98.94%, 99.74%
	[265]	HiMoReNet (Hierarchical Neural Network)	MPJPE: 64.9mm, PA-MPJPE: 46.3mm, Acceleration Error: 23.1mm/frame ²
	[266]	Motion-driven Spatial and Temporal GCN	90.8% (CS), 96.4% (CV) accuracy on NTU-RGB+D 60; 38.7% top-1 accuracy on Kinetics-Skeleton
	[267]	Multi-Stream CNN, LSTM	CC: 0.84, RMSE: 0.11, R2: 0.92 (force estimation)
	[268]	CNN	Acc: 97.5%

RNN: Recurrent Neural Network, CGAN: Conditional GAN, CNN: Convolution Neural Network, LSTM: Long Short-Term Memory, AE: Auto-Encoder, DBM: Deep Boltzmann Machines, 1DCNN: 1D Convolutional Neural Network, SAE: Sparse AutoEncoder, GCN: Graph Convolutional Network, MLP: Multi-layer Perceptron, SDAE: Stacked Denoising Autoencoder, GRU: Gated Recurrent Units, HS-CNN: Hierarchical-Split CNN, RCNN: Region-based Convolutional Neural Network, BiGRU: Bidirectional Gated Recurrent Unit, MPA: Marine Predator Algorithm, CBAM: Convolutional Block Attention Module, LH-ViT: Lightweight Hybrid Vision Transformer, HAR: Human Activity Recognition, BiLSTM: Bidirectional Long Short-Term Memory, Acc: Accuracy.

Recent advancements in gesture and motion recognition leverage sensor-based systems and ML models, such as IMU sensors [263], plastic-optical-fiber sensors [264], hierarchical neural networks such as HiMoReNet [265], graph convolutional networks for skeleton-based action recognition [266], sEMG signals for continuous gesture recognition [267], [292], and TinyDL models for air handwriting recognition [268], to improve accuracy and real-time performance in applications such as human-computer interaction, prosthetics, and handwriting recognition.

Table 10 provides a comprehensive summary of recent research work on HAR.

1) DISCUSSION AND CRITICAL INSIGHTS

HAR has seen remarkable progress driven by advances in deep learning, sensor technologies, and hybrid modeling techniques. CNNs and Long Short-Term Memory (LSTM) models continue to dominate the field due to their strong performance in extracting spatial and temporal features from wearable and embedded sensor data. Moreover, hybrid

models—such as CNN-LSTM combinations, attention-enhanced architectures, and transformer-based systems—have achieved state-of-the-art results across a wide range of HAR tasks, from daily activity detection to gesture and locomotion recognition. Emerging domains like air handwriting and EEG/EMG-based recognition also highlight the field's increasing versatility and application potential.

Despite these achievements, several limitations persist. Many studies rely on benchmark datasets that are limited in scope, size, or diversity, which raises concerns about the generalizability of reported results. Real-world deployment scenarios—such as in noisy or dynamically changing environments—are rarely simulated, and only a few studies address the robustness of models to sensor noise, placement variability, or user heterogeneity. Additionally, the use of complex models like transformers and GANs often sacrifices interpretability, making their outputs difficult to trust in high-stakes applications such as healthcare or surveillance.

Another gap lies in the limited use of multimodal and context-aware inputs. While some models integrate vision, motion, and audio data, few systems effectively fuse data from multiple sensor types in real time. The lack of standardized performance metrics and inconsistent experimental protocols further complicates cross-study comparison. Moreover, energy efficiency and latency—critical factors for mobile and embedded HAR systems—are seldom evaluated or optimized in many studies.

Future research should focus on building lightweight, explainable, and adaptive models suitable for deployment on edge devices and in resource-constrained environments. Greater emphasis should also be placed on transfer learning and self-supervised techniques to reduce dependency on labeled data, as well as privacy-preserving approaches to handle sensitive behavioral data. Finally, real-world validations, cross-cultural testing, and user-centered evaluations are needed to bridge the gap between academic performance and practical usability in diverse HAR applications.

G. TRANSPORTATION AND MOBILITY

In the domain of transportation and mobility, ML and DL techniques have been widely applied to enhance road safety, predict driver behavior, and optimize transportation systems. A primary focus has been on using predictive models and sensor data for driver behavior detection, including driver drowsiness and distraction. Models such as SVM, RF, and CNN have been used to monitor eye and mouth movements in real-time to detect drowsiness, as seen in driver fatigue detection systems [269]. IoT-based frameworks have been proposed to integrate ML models for non-intrusive real-time monitoring of driver behavior [270], [271], while transfer learning approaches utilizing VGG-16 and LightGBM were employed to track driver eye movements [272].

Driver behavior prediction models also utilize multitask learning approaches and attention mechanisms, such as in the personalized system RsSafe, which helps drivers

make safer decisions [273]. In addition, supervised learning algorithms such as RF, SVM, and LR have been applied to classify aggressive and non-aggressive driving behaviors based on sensor data [274]. Other studies have focused on detecting unsafe trucker behaviors using DL-based video analysis [275], and a multimodal system combining DL and NLP has been employed to eliminate driver distractions [276]. Meanwhile, the RoadSafety app integrated DNN and YOLO models to monitor accident risks and provide live updates to drivers [277], achieving a high detection precision, though limited in obstacle recognition.

In the context of travel behavior analysis, studies have used models such as LightGBM, XGBoost, and SVM to predict travel mode selection [278], while Bayesian models have been applied to detect pattern changes in individual travel behaviors [279]. Other studies focused on university student travel behaviors, employing SVM and RF to analyze active transportation choices, such as walking [280], [281]. Classification of traveler choices using NN also demonstrated high accuracy in predicting behavioral patterns [282].

To enhance transportation infrastructure, models such as Gradient Boosting (GB), Poisson regression, and NN have been applied to assess real-time traffic safety and monitor pedestrian behavior, providing insights for city planning [283], [284]. DNN models have also been used to predict travel behaviors with high accuracy [285], and AdaBoost algorithms classified driving behaviors with exceptional performance based on OBD data [286]. In traffic safety, studies applied Auto-Encoder (AE) models to detect anomalous driving behaviors from naturalistic driving data [287], and other work used K-means clustering to identify risky driving patterns from vehicle trajectory data [288].

Table 11 provides a comprehensive summary of recent research in transportation and mobility, detailing the contributions, algorithms used, performance metrics, and limitations of various studies focused on transportation and mobility.

1) DISCUSSION AND CRITICAL INSIGHTS

The application of ML in transportation and mobility has shown promising results across diverse tasks, including driver behavior detection, travel behavior prediction, and traffic safety optimization. Deep learning architectures such as CNNs, ResNet50, and DNNs have been effectively deployed for real-time monitoring of drowsiness, distractions, and aggressive driving, while multimodal systems leveraging vision, speech, and sensor inputs further enhance context-awareness in driver assessment. Similarly, supervised models like SVM, RF, and AdaBoost have demonstrated strong performance in classifying travel modes, driving behaviors, and transportation preferences, especially in structured environments.

However, a closer examination reveals several limitations that hinder real-world applicability. Many studies rely on datasets collected from constrained environments, such as

TABLE 11. Summary of research works on transportation and mobility.

Ref.	Contribution	Algorithms	Performance	Limitation
[277]	Accident risk prediction, landmark analysis, and driving behavior monitoring integrated into RoadSafety app for live driver updates	DNN, YOLO object detection model	MSE of 0.07565 (training) and 0.26836 (test); high detection precision for vehicles (e.g., car: 86.558%, bus, 87.054%)	Data limited to Washington D.C., the model does not monitor for some obstacles such as animals or road signs
[269]	Drowsiness detection through monitoring eye and mouth movements using CV and ML techniques in real-time	Modified SVM, KNN, RF, LR, NB	Acc: 95.20% (SVM), 93.49% (LR), 92.46% (RF), 80.82% (KNN), 69.18% (NB)	Limited performance under poor lighting conditions and requires real-time monitoring setup
[274]	Classification of driver behavior using supervised learning algorithms	LR, DT, RF, SVM, LDA	Acc: 76% (LR,SVM), 75% (LDA), 65.89% (DT)	Dataset limited to one geographic area, performance differences among classifiers under certain conditions
[279]	Detecting pattern changes in individual travel behavior using a Bayesian change-point detection model	Bayesian change-point detection	Acc: 76% (frequency dimension), 3.49% (temporal dimension), 1.38% (spatial dimension)	Some data limitations (e.g., missing values) due to privacy concerns, only private car GPS data used
[270]	Real-time detection of driver drowsiness using ML with a Haar cascade classifier and CNN	Haar cascade classifier, CNN	Acc: 97% (Haar Cascade), 96% (SVM)	Limited by the quality of the camera and environmental conditions (e.g., lighting)
[271]	IoT-based non-intrusive driver drowsiness detection framework for logistics and public transport applications	Dlib facial landmark detection, Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Euclidean Distance (ENED)	Acc: 96%	Limited by external factors such as lighting conditions and camera angles, misclassification due to reflections from spectacles
[276]	Multimodal system for detecting and eliminating driver distractions using DL	ResNet50, BiLSTMs, CNN, DeepSpeech for speech recognition	Acc: 94.1% (action classification), 95.19% (command classification); head pose estimation achieved 6.21-degree MAE	Not designed for detecting retinal abnormalities (e.g., sleeping with eyes open), may confuse similar actions such as 'talking to a passenger' and 'safe driving'
[272]	Transfer learning approach for driver drowsiness detection using eye movement behavior	VGG-16, LightGBM, KNC	Acc: 99% (KNC)	Limitations in variability of lighting conditions and potential camera quality issues
[273]	A personalized driver behavior prediction system using Multi-task Learning	MTLA, LSTM, Attention Mechanism	Acc: 96% (F-measure), ROC of 97%	Lacks integration of physiological data such as HR or stress levels
[275]	Trucker behavior security surveillance using human parsing for detecting unsafe driving behaviors based on video analysis	ResNet50, Human Parsing, CNN	Acc: 98.75% (ResNet50)	Limitations include challenges with low-light conditions and minor misclassifications due to background clutter
[282]	Classify traveler choices based on behavioral patterns using NN	NN	Acc: 92%	Performance drops for trips not on the same network as training data
[288]	Classify driving behavior using vehicle trajectory data to identify risky behaviors	K-means clustering	Acc: 85%	Exhibits poor generalization outside work zone settings owing to certain data prerequisites
[283]	Real-time traffic safety assessment utilizing connected car data, focusing on predicting crashes.	GB Regression, Poisson regression, and OLS	Best model (GB) had an R^2 of 0.806 with an RMSE of 166.36.	Outlier sensitivity and unbalanced crash data are impacting model reliability
[286]	Classification of driving behaviors with OBD data and AdaBoost algorithms	AdaBoost	Acc: 99.8%	The model does not Account for driving circumstances, including weather
[280]	ML algorithms to predict transportation choices among university students	SVM, LR, and RF	Acc: 79%	Sample limited to a specific university district, making the model not generalizable
[281]	Examine the active transportation options utilized by students	SVM and RF	Acc: 86% (SVM)	Insufficient representation of bicycle students in the sample, affecting generalizability
[278]	Explores factors affecting travel mode selection, emphasizing travel patterns and behaviors within multimodal transportation systems	SVM, LightGBM, and XGBoost	Acc: 86%	External factors such as weather and multimodal excursions were not considered in the data analysis
[284]	Monitoring pedestrian activity with ML algorithms to analyze behavior for enhancing walking infrastructure	NN	Acc: 95%	Constrained by the quality of real-time tracking data, which may fluctuate depending on sensor coverage
[285]	Evaluating DNN models alongside discrete choice models for the equitable prediction of travel behavior	DNN, LR	Average Acc of 95.71% (DNN)	DNNs are regarded as Closed box models, complicating the interpretation of predictions in a significant manner
[287]	Anomalous driving behavior detection using naturalistic driving data and AE	AE	Acc: 96.31%	Does not classify unsafe driving behaviors

DT: Decision Tree, LDA: Linear Discriminant Analysis, CNN: Convolution Neural Network, BiLSTM: Bidirectional Long Short-Term Memory, NB: Naive Bayes, KNC: K-Neighbors Classifier, RF: Random Forest, MTLA: Multi-task Learning with Attention, LSTM: Long Short-Term Memory, LR: Logistic Regression, GB: Gradient Boosting, ML: Machine Learning, OLS: Ordinary Least Squares, DNN: Deep Neural Networks, AE: Auto-Encoder, DL: Deep Learning, IoT: Internet of Things, NN: Neural Network, LightGBM: lightgradient-boosting machine, Acc: Accuracy, KNN: K-Nearest Neighbors, SVM: Support Vector Machine.

single geographic areas, specific user groups (e.g., university students or urban commuters), or synthetic settings, reducing the generalizability of findings. The variability in lighting, weather, and sensor quality—particularly for driver monitoring systems—remains a critical barrier for real-time deployment. Moreover, several models assume consistent camera positions or ideal sensor coverage, which is rarely the case in dynamic traffic scenarios.

In terms of interpretability, closed box models such as DNNs and ensemble learners often lack transparency, making it difficult to extract actionable insights or justify decisions in high-risk environments like road safety or insurance assessment. Furthermore, most existing systems

focus on classification accuracy, with limited attention paid to latency, energy consumption, and integration with existing infrastructure—factors that are vital for deployment in smart vehicles and cities.

Future research should prioritize robust, multimodal frameworks that combine sensor fusion with contextual data such as weather, time of day, and road conditions. Explainable AI approaches are essential to ensure trust and accountability in decision-making, especially in safety-critical applications. Additionally, longitudinal studies capturing real-world variability and edge-ready models optimized for low-power devices will be critical to scaling intelligent transportation systems in heterogeneous environments.

TABLE 12. Summary of financial decision-making research.

Ref.	Contribution	Algorithms	Performance	Limitations
[289]	Predicting China's crude oil futures prices by incorporating investor sentiment	LSTM	Improved price prediction Acc by 10% over traditional methods.	Limited to China's crude oil futures market
[293]	Predicting stock price crash risks by fusing multiple sentiment indicators	Ensemble Learning (Boosting, Bagging, NSGA-II)	Acc: 92.5%	Limited focus on sentiment from foreign investors
[294]	Predicting stock index prices using an integrated ensemble approach combining investor sentiment	NSGA-II	Trading simulations show a 15% improvement in profitability	Limited scope of market datasets
[291]	Hybrid model combining DL and investor sentiment for stock price prediction	CNN, LSTM	Improved prediction Acc by 8.3% with sentiment and technical indicators	Limited datasets for validation
[297]	Credit scoring using PCA combined with LR	PCA, LR	Increased credit score prediction Acc by 12%	Dependent on the quality of input financial data
[301]	Predicting individual stock prices by integrating investor sentiment analysis with BERT and BiGRU	BERT, BiGRU, Attention Mechanism	Acc: 89.3%	Focused on Chinese financial markets
[302]	Using ML techniques for fraud detection in the Accounting and finance sector	SVM, AE, RF, CNN	Acc: 95%	Limited evaluation in sectors beyond Accounting
[310]	Stock market prediction model combining ARIMA, LSTM, and LR with emotional analysis	ARIMA, LSTM, LR	Improved stock prediction Acc by 7.4%	Limited testing across broader stock markets
[306]	Ontology-based model for credit scoring in microfinance	Ontology-based Modeling	×	Limited real-world validation of the ontology model
[295]	Using ML and sentiment analysis to predict the Moroccan stock market's movements	Sentiment Analysis, ML (RF, SVM)	Improved stock price prediction Acc by 9.2%.	Limited to the Moroccan stock market
[300]	Credit score prediction using SVM optimized with GWO	SVM, GWO	Improved credit score prediction Acc by 11.3%.	Limited evaluation in industries beyond banking
[298]	Using AI and ML for risk assessment in lending to evaluate creditworthiness	AI, ML	Acc increased by 13.7% in predicting creditworthiness	Limited real-time data usage
[303]	Predicting financial risks and anomalies in enterprises using ARIMA and DL techniques	ARIMA, DL	Improved financial risk prediction by 14.8%	Limited scope of validation datasets
[305]	Bank credit risk assessment using BP NN	BP NN	Acc: 92.58%	Limited testing across diverse financial institutions
[299]	Credit-risk model combining SVM, DT, NN, and LR	SVM, DT, NN, LR	Improved credit risk prediction by 10%	Requires high-quality data
[304]	Credit score model using a fusion NN combining CNN and RNN	CNN, RNN	Acc: 85.3%	Limited evaluation in real-world financial sectors
[307]	Financial loan risk assessment using fuzzy cluster analysis	Fuzzy Cluster Analysis	Improved loan risk assessment Acc by 9.5%.	Focuses on specific financial sectors
[308]	Improving cross-selling prediction for consumer loans using DL	AE	AUC improved by 4-5.4% in cross-selling predictions	Limited to consumer loans
[309]	Stock market prediction using sentiment analysis combined with ML techniques such as SVM and ANN.	SVM, ANN	Prediction Acc improved by 8.9%	Limited testing across broader financial markets
[296]	Stock market price prediction using news sentiment analysis and ensemble learning	Ensemble Learning	Improved stock price prediction by 7.2%	Limited to specific stock markets

RNN: Recurrent Neural Networks, AI: Artificial Intelligence, CNN: Convolution Neural Network, PCA: Principal Component Analysis, DL: Deep Learning, LR: Logistic Regression, BiGRU: Bidirectional Gated Recurrent Unit, LSTM: Long Short-Term Memory, SVM: Support Vector Machine, ANN: Artificial Neural Network, AE: Auto-Encoder, ARIMA: AutoRegressive Integrated Moving Average, ML: Machine Learning, NN: Neural Network, ×: Not Specified, RF: Random Forest, GWO: Gray Wolf Optimization.

H. FINANCIAL DECISION-MAKING

In financial decision-making, various ML and DL models have been applied to enhance prediction accuracy and improve risk assessment. Stock market prediction has been a key focus, with models such as LSTM and CNN integrating investor sentiment and technical indicators to predict stock prices and crude oil futures, significantly improving prediction accuracy [289], [291]. Ensemble learning techniques combined with sentiment analysis have been applied to predict stock price crashes and market movements, demonstrating improved robustness [293], [294], [295], [296].

For credit scoring and financial risk assessment, hybrid models incorporating techniques such as PCA, SVM, and DT have been explored to enhance prediction accuracy, with applications in creditworthiness evaluation and lending processes, showing improved risk assessment [297], [298], [299], [300]. BERT, BiGRU, and attention mechanisms were used to predict individual stock prices, combining sentiment data for high-accuracy predictions [301].

Risk management and fraud detection have seen improvement through ML models such as SVM, RF, and ARIMA. Also DL models which were applied to detect

fraudulent activities and manage financial risks in enterprises [302], [303], [304], [305]. Fuzzy cluster analysis and ontology-based models were also used in credit risk assessment, particularly in microfinance, offering more nuanced risk evaluation frameworks [306], [307].

In cross-selling prediction for consumer loans, DL models such as Auto-Encoder (AE) have shown improvements in prediction accuracy, particularly in consumer-focused financial services [308]. Studies using SVM, ANN, and sentiment analysis demonstrated better performance in predicting stock market trends [309]. Additionally, hybrid models combining ARIMA, LSTM, and linear regression were used to forecast stock market prices, utilizing sentiment analysis from social media [310].

Table 12 provides a summary of recent research contributions to financial decision-making, highlighting the algorithms used, key performance metrics, and any notable limitations.

1) DISCUSSION AND CRITICAL INSIGHTS

ML and DL have significantly transformed financial decision-making processes, particularly in areas such as stock

market forecasting, credit scoring, fraud detection, and risk assessment. LSTM, CNN, and ensemble models have shown consistent success in modeling non-linear financial data patterns, especially when combined with sentiment analysis. These models have enhanced the prediction of stock price movements, crash risks, and individual creditworthiness, while hybrid approaches—such as integrating PCA or optimization algorithms like GWO—have improved accuracy and robustness.

Despite these advances, several limitations persist. Many studies are constrained to specific markets (e.g., Chinese or Moroccan financial systems), which raises concerns about generalizability to other economic contexts. Additionally, the reliance on historical financial data or static sentiment indicators means that models often fail to adapt to volatile or highly dynamic conditions. While some models show high accuracy in experimental settings, their performance in real-world, live trading or lending environments remains under-explored.

Interpretability is another key issue, particularly with deep and ensemble models. These models often operate as “closed boxes,” making it difficult for financial analysts or regulators to validate predictions or understand underlying decision mechanisms. This is particularly problematic in domains such as credit scoring and fraud detection, where transparency and fairness are critical.

Moreover, few studies incorporate real-time or streaming data, limiting their effectiveness in high-frequency trading or live credit risk management. Ethical considerations—such as model bias, discrimination in lending, and data privacy—are rarely addressed in technical evaluations, despite their importance in financial AI systems.

To advance this field, future work should prioritize explainable AI approaches, real-time data integration, and robust testing across diverse markets. There is also a need to establish standardized benchmarks and cross-market datasets for comparative evaluation. Finally, interdisciplinary collaboration with economists, regulatory bodies, and social scientists will be crucial to ensure ethical, equitable, and effective deployment of AI in financial decision-making.

VI. CHALLENGES AND LIMITATIONS IN HBA

While HBA has advanced significantly with the integration of AI, ML, and data science, several challenges and limitations persist. These challenges are crucial to address to ensure the effective, ethical, and accurate application of HBA in real-world scenarios. This section outlines key challenges related to data privacy, generalization, real-time processing, and scalability

A. DATA PRIVACY AND ETHICAL CONCERNS

One of the most pressing challenges in HBA is ensuring *data privacy* and addressing *ethical concerns*. Since HBA systems analyze personal and sometimes highly sensitive information—such as physiological signals, social media

activity, and geolocation data—protecting individuals’ privacy is paramount. The collection, storage, and processing of such data raise significant ethical dilemmas, including the necessity of *informed consent*, *anonymization*, *security*, and the potential for *misuse* [311], [312]. Below are the key aspects of these concerns:

1) INFORMED CONSENT AND ANONYMIZATION

Ensuring that individuals provide explicit, informed consent before their behavioral data is collected and analyzed is fundamental. However, obtaining meaningful consent in large-scale HBA systems, particularly those that operate passively in the background—such as smart surveillance systems or wearable IoT devices—remains an ongoing challenge [313], [314]. Often, users are unaware that their data is being collected, making true informed consent difficult to achieve. Even when anonymization techniques are applied, the risk of re-identification persists [315], [316]. Combining multimodal data sources, such as facial recognition and geolocation tracking, can inadvertently reveal identities even if personally identifiable information (PII) has been removed. This calls for stronger data protection measures and clearer communication between data collectors and users regarding how their data will be used.

2) DATA SECURITY AND PROTECTION AGAINST BREACHES

As HBA systems handle large volumes of sensitive data, they become lucrative targets for cybercriminals. Unauthorized access to behavioral data can lead to serious consequences, including identity theft, financial fraud, and intrusive surveillance [317]. Securing this data requires robust encryption, access control mechanisms, and continuous monitoring to detect and mitigate security threats [318]. However, achieving this level of security is complex, as behavioral data is often distributed across multiple devices and cloud-based systems. Additionally, the increasing reliance on IoT devices introduces further vulnerabilities, as many of these devices lack advanced security features. Without proper cybersecurity measures, HBA systems risk exposing user data to malicious entities [319].

3) ETHICAL USE AND POTENTIAL FOR MISUSE

While HBA has the potential to enhance various domains, including healthcare and security, its use also raises ethical dilemmas. Behavioral data can be repurposed for applications beyond its original intent, sometimes without user awareness or consent. For instance, companies collecting user interaction data for personalized recommendations could later use it for targeted advertising or even behavioral manipulation [320]. Similarly, governments or organizations could deploy HBA for mass surveillance [321], raising concerns about civil liberties and personal freedoms. Establishing clear regulatory frameworks and ethical guidelines is essential to ensure that HBA is used responsibly, without infringing on individuals’ rights.

4) BIAS AND DISCRIMINATION IN HBA MODELS

AI models used in HBA are only as good as the data they are trained on. If the training data is not diverse, the resulting models may exhibit bias, leading to discriminatory outcomes. For example, an emotion recognition system trained on data from one cultural group may misinterpret expressions from individuals belonging to different ethnic backgrounds [322]. Similarly, bias in hiring assessments powered by HBA could lead to unfair treatment of candidates based on gender, race, or socioeconomic status. Addressing these biases requires the use of diverse datasets, fairness-aware algorithms, and continuous auditing to identify and mitigate discriminatory patterns in AI models.

B. GENERALIZATION AND BIAS ISSUES

HBA models often struggle with *generalization*, meaning that they may perform well on specific datasets but fail when applied to different populations, cultural settings, or real-world scenarios. Several factors contribute to this limitation:

1) LIMITED GENERALIZATION ACROSS DEMOGRAPHICS

Human behavior is influenced by a variety of factors, including cultural norms, social upbringing, and environmental conditions. However, many HBA models are trained on datasets that reflect only a subset of the global population. For instance, a facial expression analysis model developed using Western datasets may not accurately interpret emotions expressed by individuals from Asian or African cultures. This lack of generalization results in reduced accuracy and potential misinterpretations when the model is applied to a broader audience [323], [324].

2) BIAS IN DATA COLLECTION

Biases often emerge at the data collection stage, particularly when certain groups are overrepresented or underrepresented in the dataset. If the majority of the training data comes from urban populations, an HBA model may struggle to analyze behaviors in rural settings. Similarly, if behavioral data is collected primarily from young adults, the model may perform poorly when assessing elderly individuals [325]. Ensuring balanced, diverse datasets that capture the full spectrum of human behavior is crucial for improving the generalizability of HBA models.

3) ADAPTING TO NEW CONTEXTS AND ENVIRONMENTS

HBA models often fail when deployed in environments different from those they were trained in. A system designed to monitor stress levels in office workers may not perform effectively in a high-intensity hospital setting, where behavioral cues differ. One way to address this is through techniques such as transfer learning and domain adaptation, which allow models to adjust to new datasets with minimal retraining [326].

C. REAL-TIME PROCESSING AND SCALABILITY

For many HBA applications, *real-time processing* is essential to ensure timely responses and decision-making [327]. However, several technical challenges hinder the real-time capabilities and scalability of HBA systems:

1) HIGH COMPUTATIONAL REQUIREMENTS

Analyzing behavioral data in real time requires significant computational power, particularly when dealing with multimodal inputs such as video, audio, and sensor data. Processing these streams efficiently without introducing delays remains a challenge. Traditional cloud-based architectures often introduce latency, making them unsuitable for time-sensitive applications like emergency response systems or real-time emotion detection [328].

2) SCALABILITY ACROSS DEVICES AND PLATFORMS

HBA models must function consistently across various devices, ranging from high-performance cloud servers to resource-constrained edge devices such as smartphones and IoT sensors. Ensuring seamless performance across these platforms requires optimized algorithms that can adapt to different hardware capabilities. Edge computing and lightweight deep learning models offer potential solutions, enabling real-time processing on-device while minimizing reliance on cloud infrastructure [329], [330].

3) MANAGING BANDWIDTH AND STORAGE CONSTRAINTS

Continuous data collection from multiple sources generates vast amounts of information, leading to high storage and bandwidth demands. Real-time streaming of behavioral data can quickly overwhelm network resources, particularly in large-scale deployments. Implementing efficient data compression [331], feature extraction, and selective data transmission strategies [332] can help mitigate these issues while maintaining system performance.

4) LATENCY AND SYSTEM DELAYS

High-latency systems undermine the effectiveness of real-time HBA applications. For instance, a healthcare monitoring system that detects anomalies in patient behavior must provide instant alerts to caregivers to be useful. Optimizing processing pipelines, using dedicated AI accelerators, and leveraging parallel computing techniques are crucial for minimizing delays and ensuring real-time responsiveness [333].

VII. EMERGING TECHNOLOGIES AND EXPANDING METHODOLOGICAL HORIZONS IN HBA

HBA continues to evolve rapidly in response to novel technologies and data-centric paradigms that are redefining how behavior is perceived, captured, and interpreted. This section synthesizes emerging technological trends and

methodological advances that are reshaping the field, with a focus on practical implementation, contextual awareness, and system-level integration.

A. EDGE COMPUTING FOR LOW-LATENCY, CONTEXT-AWARE INFERENCE

Edge computing is becoming indispensable in HBA applications where latency, privacy, and connectivity constraints pose significant concerns [335]. By moving computation from cloud servers to local devices—such as smartphones, wearables, and embedded IoT nodes—edge systems enable real-time processing of behavioral signals like posture, speech, and physiological indicators. This architecture supports responsive environments, such as ambient-assisted living [336] and driver monitoring systems [10], [337], where milliseconds of delay can affect safety or decision relevance.

Notable implementations include emotion recognition engines running on microcontrollers and wearable stress monitoring that infer behavior without cloud dependency. These systems utilize lightweight models optimized via pruning, quantization, or TinyML frameworks [339] to achieve near real-time performance. For instance, adaptive offloading strategies [338] have been proposed where data is selectively transmitted to the cloud only when local confidence thresholds are not met, balancing energy consumption and inference accuracy. The shift toward edge-native behavioral analytics is also contributing to stronger data localization and regulatory compliance in privacy-sensitive domains.

B. AI-DRIVEN BEHAVIORAL MODELING WITH COGNITIVE DEPTH

As ML models become increasingly sophisticated, there is a move toward capturing not just observable actions but also internal cognitive-emotional states and social signals. Deep learning architectures now support multi-layered understanding of human behavior that reflects intention, affect, and context. For example, reinforcement learning has been leveraged to simulate behavioral adaptation in tutoring systems and social robotics, learning optimal feedback policies through interaction rather than static labels [334], [340].

Transformer-based networks, which have revolutionized natural language processing, are being adapted for spatiotemporal action recognition and multi-agent behavior modeling. These models offer greater contextual memory and support for long-range dependencies, which are essential for detecting subtle shifts in behavior over time [341]. In addition, neural-symbolic approaches are being explored to integrate statistical learning with rule-based reasoning, enabling systems to incorporate domain knowledge and causality into behavioral inference. These directions are enabling behavior models to be not just predictive but explanatory and interactive [342].

C. ADVANCED SENSING MODALITIES AND MULTIMODAL INTEGRATION

Sensor advancements are enhancing the fidelity and granularity of behavioral capture. Traditional sensors—accelerometers, gyroscopes, and RGB cameras—are now complemented by biosensors (e.g., skin temperature, electrodermal activity), LiDAR for depth-based gesture recognition [345], and near-infrared cameras for driver fatigue detection in low light. These modalities are often used in combination to build rich, multimodal datasets that capture not just motion but context and affect.

Recent studies have integrated real-time EEG and ECG analysis with facial micro-expression detection for applications in emotion-aware systems and cognitive workload monitoring [343]. Wearable patches and smart garments are also being deployed to unobtrusively monitor posture and stress in workplace settings. Importantly, multimodal fusion strategies—such as early fusion with shared encoders, late fusion with ensemble voting, and hybrid attention mechanisms—are proving effective in synchronizing heterogeneous inputs [344]. These approaches enhance system robustness and allow for adaptive feature selection depending on environmental constraints or task demands.

D. LEVERAGING LARGE LANGUAGE MODELS FOR TEXTUAL BEHAVIORAL INSIGHTS

Large Language Models (LLMs) such as BERT, GPT, and T5 are transforming textual behavior analysis by enabling semantic understanding of user-generated content across emails, social media, chat logs, and transcriptions [346], [347]. Unlike rule-based NLP systems, LLMs can infer user intent, detect sentiment drift, and recognize psychosocial markers of distress or engagement. In mental health monitoring, for instance, LLMs have been used to detect depressive tendencies from Reddit posts or analyze therapeutic conversations for empathy and coherence.

More advanced implementations include dialogue agents that adapt their conversational strategies based on real-time feedback and inferred user mood. In financial behavior prediction, LLMs have been fine-tuned to extract decision drivers from investor sentiment or customer complaints, thereby augmenting classical quantitative models [348]. Fine-tuning and prompt engineering are critical tools for customizing LLMs to behavioral tasks while preserving general linguistic competence. Importantly, when combined with other modalities—such as vision or physiology—LLMs serve as the linguistic layer of multimodal HBA systems, bridging narrative and numerical reasoning.

E. ETHICAL AND PRIVACY-PRESERVING TECHNIQUES

As HBA technologies become increasingly pervasive in healthcare, smart cities, education [349], and workplace monitoring, the need to protect user data while maintaining the integrity and fairness of analytical models has never been greater. Solutions to these concerns go beyond

high-level principles, increasingly incorporating concrete technical methods and governance strategies. This subsection explores multiple privacy-preserving and ethically grounded approaches that are shaping the responsible evolution of HBA [350].

1) DIFFERENTIAL PRIVACY

Differential privacy is a formalized approach to protecting individual data contributions by adding controlled statistical noise to outputs or intermediate computations [351]. This technique ensures that the presence or absence of a single data point does not significantly alter the outcome of the analysis, thus maintaining plausible deniability for individuals. For HBA tasks such as behavioral clustering or emotional state prediction on social media data, researchers can apply differentially private mechanisms at the data aggregation level or during model training [352]. Google's RAPPOR and Apple's iOS telemetry systems are real-world deployments of this technique. Tools such as TensorFlow Privacy and PyTorch Opacus allow researchers to build differentially private deep learning models that can be applied in domains like mental health analytics [353] or educational feedback without exposing individual identities.

2) FEDERATED LEARNING

Federated Learning (FL) [354] is a decentralized ML paradigm that allows multiple devices or silos (e.g., hospitals, smartphones, smartwatches) to collaboratively train a shared model while keeping raw data locally stored. FL is ideal for behavior recognition using physiological data collected from wearables, enabling privacy-preserving applications in stress detection, fitness monitoring, or early disease warning [355]. To enhance FL's robustness, recent advancements such as secure aggregation protocols, differential privacy integration, and homomorphic encryption are being explored to prevent leakage through gradient updates [356]. In HBA systems, FL can also support personalization by adapting global models to local behavior patterns via fine-tuning or federated meta-learning techniques [357].

3) SECURE MULTI-PARTY COMPUTATION (SMPC)

Secure Multi-party Computation (SMPC) allows multiple entities to compute a joint function over their inputs without revealing the inputs themselves. This cryptographic approach [358] is increasingly being investigated in scenarios like collaborative behavioral research across hospitals or academic institutions. For example, SMPC enables multiple mental health clinics to contribute to a shared stress prediction model without disclosing individual patient records, thereby supporting both privacy and research goals.

4) SYNTHETIC DATA GENERATION

In cases where real data cannot be shared or is insufficient, synthetic data can be generated using methods such as Generative Adversarial Networks (GANs) [359].

These synthetic datasets preserve statistical properties of the original data while removing personal identifiers. In HBA, this technique is particularly useful for training models on sensitive behaviors such as addiction relapse or employee burnout. By simulating human-like behavior, these datasets enable safe experimentation and benchmarking.

5) ETHICAL AI FRAMEWORKS

Frameworks such as the EU's High-Level Expert Group on AI Ethics Guidelines [360], IEEE's Ethically Aligned Design, and the Montreal Declaration on Responsible AI provide practical checklists and design principles. These include guidelines for fairness, explainability, and non-discrimination. In the context of HBA, adopting these frameworks translates into deploying bias mitigation strategies (e.g., reweighting, adversarial debiasing), using interpretable models (e.g., SHAP, LIME), and conducting algorithmic audits to verify compliance [361].

6) CONSENT MANAGEMENT PLATFORMS (CMPs)

Modern HBA systems, especially those embedded in mobile apps or online services, are integrating CMPs to facilitate transparent data governance. These platforms allow users to opt in or out of specific data usage scenarios and provide fine-grained control over consent types (e.g., for training vs. personalization). Coupled with real-time dashboards, CMPs empower users to make informed choices about their behavioral data [362].

7) DATA MINIMIZATION AND PURPOSE LIMITATION

HBA platforms are also being designed to implement the principles of data minimization (collecting only the data necessary for a task) and purpose limitation (restricting data use to its stated objectives). Techniques such as feature selection, task-specific embeddings [363], and model pruning [364] are used to avoid unnecessary collection of behavioral signals. This reduces the attack surface for privacy breaches while improving model efficiency.

8) ETHICAL OVERSIGHT COMMITTEES AND IMPACT ASSESSMENTS

Beyond technical tools, organizations deploying HBA at scale are instituting AI ethics boards and Data Protection Impact Assessments (DPIAs) [365]. These structures enable systematic evaluations of behavioral data usage, especially in high-stakes areas like law enforcement or education. DPIAs help in identifying and mitigating risks before deployment, ensuring regulatory compliance (e.g., GDPR) and ethical accountability.

F. CROSS-DISCIPLINARY APPROACHES

The future of HBA is likely to involve an increasing integration of cross-disciplinary approaches, combining insights from various fields to drive innovation. Key areas where cross-disciplinary approaches are making an impact include:

1) PSYCHOLOGY AND AI

The integration of psychological theories with AI models has proven instrumental in constructing more nuanced and interpretable behavioral systems. Concepts such as cognitive dissonance, behavioral reinforcement, emotional regulation, and attention mechanisms inform the architecture and function of ML models. For instance, cognitive-behavioral models [366] can be used to shape personalized feedback mechanisms in intelligent tutoring systems or mental health applications. Emotion recognition systems often draw from established psychological taxonomies such as Ekman's six basic emotions or Russell's circumplex model [367], enabling algorithms to ground prediction in empirically validated frameworks. Psychological profiling also informs recommendation systems, where user traits like openness or impulsivity are used to predict preferences or behavioral risk. By embedding these theories into algorithmic design, HBA systems become more aligned with real-world human variability and less prone to oversimplification or bias.

2) NEUROSCIENCE AND BEHAVIORAL MONITORING

Advances in neuroscience are allowing HBA systems to go beyond observable behavior and tap into the underlying neural correlates of cognition and emotion. Wearable neuroimaging tools such as EEG and functional near-infrared spectroscopy (fNIRS) [368] offer real-time access to brain activity in naturalistic settings. These data can be used to train multimodal models that associate neural patterns with behavioral responses, enabling more accurate detection of cognitive load, emotional arousal, or attention lapses. For example, combining neural signals with facial expressions and speech allows for richer emotion modeling in therapeutic or educational settings [369]. In autonomous driving or high-stress environments, real-time brain monitoring can trigger adaptive system behaviors to reduce overload [370]. This fusion of behavioral and neurophysiological signals supports more precise and responsive HBA interventions, particularly in neurodivergent care or cognitive rehabilitation.

3) HEALTHCARE AND WEARABLE TECHNOLOGY

The convergence of wearable technology and healthcare has unlocked new frontiers in behavioral monitoring. Devices such as smartwatches, fitness bands, and biosensor patches continuously collect physiological indicators—heart rate, electrodermal activity, movement patterns—that correlate with psychological states such as stress, fatigue, or depression. When these data streams are interpreted alongside self-reported measures or environmental context, they allow for the creation of personalized health profiles and just-in-time adaptive interventions (JITAI) [371]. For example, in post-operative care or chronic illness management, behavioral deviations from baseline can prompt automated alerts or caregiver notifications. Collaboration between clinicians, biomedical engineers, and behavioral scientists ensures that these technologies are both medically valid and user-friendly.

Importantly, this integration also helps validate the behavioral insights produced by AI models against clinical gold standards, thereby increasing reliability and trustworthiness.

4) SOCIOLOGY AND SOCIAL BEHAVIOR ANALYSIS

Sociological theory contributes a macro-level lens to HBA by emphasizing the role of social context, group behavior, and systemic structures. Techniques such as social network analysis (SNA) [372], role theory, and symbolic interactionism provide models for understanding how individuals behave in relation to others. In online platforms, for example, sociologically informed algorithms can detect emergent phenomena like echo chambers, opinion polarization, or collective mood shifts. In organizational settings, behavior analytics grounded in social role expectations and interaction patterns can improve team cohesion, detect burnout, or forecast attrition. Tools such as dynamic network graphs [373] and sentiment-flow models are used to quantify interactions over time, revealing insights into leadership influence, information diffusion, or community resilience. By embedding sociological constructs into HBA systems, researchers and practitioners can better assess the interplay between individual behavior and collective norms, yielding insights that are both predictive and explanatory.

VIII. CONCLUSION

This survey has provided a comprehensive analysis of the current state of HBA, encompassing foundational concepts, recent computational techniques, and diverse application domains such as healthcare, education, marketing, security, and workplace well-being. By synthesizing developments in machine learning, deep learning, computer vision, and natural language processing, the review underscores how these technologies are reshaping our ability to model, predict, and interpret human behavior across real-world contexts.

Despite its comprehensive scope, the review has several limitations that must be acknowledged. First, although we consulted major academic databases, some relevant grey literature, domain-specific repositories, or emerging preprints may have been excluded. Second, the scope was limited to English-language publications, potentially omitting significant contributions in other languages. Third, the review reflects the state of the literature only up to March 2024, which means recent advances may not be captured. Additionally, the heterogeneity in methodologies, datasets, and evaluation metrics across studies makes direct performance comparisons challenging. Finally, due to these inconsistencies, we did not conduct a formal meta-analysis, which might have further quantified trends and model effectiveness.

Looking ahead, we recommend several future research directions to strengthen the field of HBA. There is a pressing need for standardized benchmarks, including publicly available datasets and unified evaluation protocols, to enable reproducibility and fair model comparison. Interdisciplinary collaboration should also be encouraged, bringing together

expertise from artificial intelligence, psychology, ethics, and social sciences to develop more holistic and socially responsible behavior analysis systems. Privacy-preserving techniques such as federated learning and differential privacy must be prioritized to ensure ethical deployment, especially in sensitive domains. Furthermore, more longitudinal and cross-cultural studies are essential to capture behavioral diversity and mitigate algorithmic bias. Future models should emphasize explainability to promote transparency in decision-making processes. Finally, advancing lightweight and real-time solutions suitable for edge computing will be critical for deploying HBA technologies in mobile, wearable, or resource-constrained environments.

In conclusion, HBA is a rapidly evolving field at the intersection of data science and human-centered disciplines. This survey aims to serve as a foundation for future exploration, offering a panoramic view of existing techniques, challenges, and opportunities. As computational tools become more integrated into daily life, the responsible and effective analysis of human behavior will play a vital role in shaping intelligent, inclusive, and adaptive systems.

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